

Technology-Driven Market Concentration through Idea Allocation*

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Abstract

This paper documents that periods of greater technological breakthrough intensity coincide with a larger startup share of innovation and lower market concentration. To explain this, we develop a quantitative-theoretic model where idea holders choose between launching startups and joining incumbents, yielding two predictions: (1) startup entry rises during technology-wave peaks, and (2) within incumbents, idea quality assortatively matches with firm size. Both predictions match the data and explain declining concentration during breakthroughs. Quantitatively, the breakthrough slowdown predicts faster average firm-quality growth, but this is offset by slower net entry. This slowdown largely accounts for market concentration's upward trend and fluctuations.

Keywords: technology waves, growth, HHI, startups, incumbent firms.

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1 Introduction

Technological breakthroughs create opportunities for growth, but their economic consequences depend on where new ideas are developed. Some ideas give rise to independent businesses, while others expand the productive capabilities of established firms. What determines this organizational allocation, and how does it change as technologies advance? This paper studies how variation in the intensity of radical innovation shapes startup formation, incumbent upgrading, and the distribution of firm size. Market concentration emerges as an endogenous outcome of this process, revealing how technological opportunities are distributed across firms.

We approach this question through the benefits and costs of developing ideas within existing firm boundaries. Incumbents provide established assets and commercialization capacity, but incorporating an idea into an existing organization also requires adapting it to legacy technologies and motivating the inventor responsible for its development. Independent startups offer a different combination of technological flexibility, commercialization resources, and incentives. Changes in technological opportunities can therefore alter the relative advantages of these organizational forms, reallocating innovation between entrants and incumbents and across incumbent firms of different sizes.

Using USPTO, Longitudinal Business Database (LBD), and Compustat data, we document five patterns linking technology waves to organizational allocation:

1. Aggregate breakthrough intensity exhibits a declining trend punctuated by medium-run waves.
2. The allocation of innovation between startups and incumbents tracks these waves: at their peaks, a larger share of patents belongs to young firms.
3. Among incumbent firms, patent quality, measured by forward citations, is positively associated with firm size.
4. Conditional on scientific value, patents have higher economic value at larger firms, while the economic return to scientific value declines with aggregate breakthrough intensity. Equity-based compensation per employee decreases with firm size, and higher lagged compensation predicts greater subsequent total patent value.
5. Market concentration, measured by employment-, payroll-, and sales-based HHIs, rises over time and fluctuates inversely with technology waves.

These patterns provide suggestive evidence that technological opportunities and organizational trade-offs shape where innovation occurs.

We develop a model in which inventors hold ideas of heterogeneous quality and technological waves shift the distribution of those ideas. For ideas requiring their continued involvement, the inventors choose between establishing a new firm with a partner and developing the idea within an incumbent of a chosen size. Startup innovation introduces new varieties, while incumbent innovation improves existing varieties. Our focus is on how technological opportunities and organizational trade-offs jointly determine the allocation of ideas across these innovation channels and firm sizes.

The model formalizes two costs of developing ideas within existing firm boundaries, alongside the commercialization benefits of incumbent scale. The first is technological integration. An incumbent must combine a new idea with its legacy technology base, limiting the extent to which improvements in idea quality translate into realized innovation value. These adoption frictions, in the spirit of [Jovanovic and Rousseau \(2014\)](#), become more consequential for frontier-shifting ideas. Startups can develop such ideas without integrating them into an existing product line, but forgo incumbents' commercialization synergies.

The second cost is the agency cost. Inventor effort is unobservable, creating a principal-agent problem between firms and inventors. Firms face a trade-off between incentives and insurance under wage-plus-equity contracts. Equity exposes inventors to both the returns from their innovations and risks associated with the firm's existing operations. As firm size increases, these risks become more important relative to the inventor's contribution, making strong incentives more costly to provide. This mechanism follows the organizational perspective of [Holmstrom \(1989\)](#) and is consistent with our suggestive evidence on firm size, equity compensation, and subsequent innovation.

The two frictions play distinct roles. Adoption frictions primarily shape the choice between startups and incumbents by making independent development relatively attractive for radical ideas. Incentive problems shape the choice among incumbents by counterbalancing commercialization synergies: without a countervailing cost of size, those synergies would direct all incumbent-bound ideas toward the largest available firms. Together, these frictions govern both whether inventors launch startups and which incumbents they join.

The model delivers two main predictions. First, at the peaks of technology waves, a larger share of inventors launch new firms to develop their ideas. Second, among inventors who conduct R&D in incumbent firms, there is positive assortative matching between idea quality and firm size: larger firms attract higher-quality ideas. Both predictions align with the data and jointly contribute to lower concentration near technology-wave peaks. The surge in startup formation expands the mass of firms, and

because startups are not subject to positive assortative matching, they counterbalance the tendency of large incumbents to accumulate more valuable ideas and expand further.

To quantify how technology waves affect growth and market concentration through idea allocation, we simulate the calibrated model annually from 1984 under two specifications. The baseline varies both the idea-quality scale parameter and the existing technology base; the second varies only the former, holding all other parameters fixed. We decompose aggregate growth into growth in average firm quality and firm mass, and compare simulated concentration and idea allocation with their empirical counterparts.

The baseline simulation generates relatively strong growth in average firm quality alongside a contraction in firm mass during 2000–2010. It also reproduces the upward concentration trend, with detrended HHI correlations of 0.552 and 0.835 with employment- and sales-based concentration, respectively. Varying only the idea-quality scale parameter generates similar growth patterns and much of the baseline concentration trend and comovement. We also compare simulated relationships with instrumental-variable estimates relating technological breakthrough intensity to industry-level concentration and the probability that a patent belongs to a young firm. The simulations reproduce the signs of both relationships: the negative concentration coefficients are approximately one-third to one-half of the empirical estimates across contemporaneous and lagged specifications, while the positive coefficient for the young-firm probability is 0.281, compared with empirical estimates of 0.127–0.179.

Idea allocation affects concentration through both extensive and intensive margins. On the extensive margin, technological breakthroughs encourage startup formation, expanding firm mass. The detrended baseline new-to-incumbent ratio correlates 0.711 with its empirical counterpart, although its decline and volatility are overstated. To isolate the intensive margin, we construct a counterfactual that preserves baseline entry and firm mass but reallocates startup-generated quality improvements to incumbents according to inventors' optimal incumbent-size choices. This reassignment raises average HHI by 11% and its trend slope from 0.013 to 0.017. Thus, startup formation reduces concentration through two channels: increasing firm mass and weakening the positive assortative matching that directs better ideas toward larger incumbents.

Related Literature

This paper connects to the endogenous growth literature on variety expansion [Romer \(1990\)](#), quality improvement [Aghion and Howitt \(1990\)](#); [Grossman and Helpman \(1991\)](#), and innovation-driven firm dynamics [Klette and Kortum \(2004\)](#); [Akcigit and Kerr \(2018\)](#). Frameworks combining multiple innovation channels, including [Howitt \(1999\)](#); [Garcia-Macia, Hsieh and Klenow \(2019\)](#), examine their contributions to aggregate growth. Our

focus on organizational allocation connects to [Chatterjee and Rossi-Hansberg \(2012\)](#), who link private information about idea quality to spinoff formation; [Ma \(2026\)](#), who studies specialization between innovation and production across firms; and [Yang \(2026\)](#), who examines how R&D ownership shapes inventor incentives. This paper examines how technological waves, adoption frictions, and incentive provision jointly determine whether ideas generate new firms or improve incumbent ones, and which incumbents attract them. These choices shape both growth and the firm-size distribution.¹

This paper explores the macroeconomic implications of groundbreaking technologies. Empirically, [Bowen III, Frésard and Hoberg \(2023\)](#) show that startups increasingly remain independent when technologies evolve rapidly, while [Dinlersoz, Dogan and Zolas \(2024\)](#) document a surge in AI business applications after 2016. [Crouzet et al. \(2026\)](#) show that young firms have a growth advantage when new technologies require organizational workstyles that conflict with established business models, highlighting organizational barriers to adoption. On the theoretical front, [Greenwood and Yorukoglu \(1997\)](#); [Atkeson and Kehoe \(2007\)](#); [Jovanovic and Rousseau \(2014\)](#) examine adaptation frictions and their aggregate growth implications, while [Comin and Gertler \(2006\)](#) link endogenous innovation and technology adoption to medium-term business cycles. We contribute to this literature by examining how technology waves interact with organizational constraints to shape startup formation, incumbent expansion, and medium-term concentration dynamics.

This paper helps explain rising U.S. market concentration since the early 2000s. While this trend has coincided with gains in allocative efficiency and productivity ([Autor et al. \(2020\)](#); [Ganapati \(2021\)](#)), it has also been accompanied by declining firm entry and a widening gap between large and small firms ([Akcigit and Ates \(2023\)](#); [Olmstead-Rumsey \(2019\)](#)), and growing innovation difficulty ([Bloom et al. \(2020\)](#)). Relatedly, [Berlingieri et al. \(2024\)](#) show how bursts of product innovation shape firm growth and the concentration of production among large firms. We emphasize how changes in technological opportunities alter the allocation of ideas between startups and incumbents. Our technology wave measures reveal major innovation peaks in the mid-1980s and mid-1990s, followed by a prolonged lull until the early 2010s. During this period, more inventors turned to incumbent firms, enabling large incumbents to expand further.

The rest of the paper is organized as follows. Section 2 introduces measures of the technology waves, the allocation of ideas, and market concentration, and subsequently presents their patterns. Section 3 constructs a model where inventors choose between initiating new ventures and joining established incumbents of a chosen size. We derive

¹Related work on innovation, firm dynamics, and growth includes [Perla, Tonetti and Waugh \(2021\)](#); [Fons-Rosen, Roldan-Blanco and Schmitz \(2021\)](#); [Hopenhayn, Neira and Singhania \(2022\)](#); [Akcigit, Celik and Greenwood \(2016\)](#); [Cavenaile, Celik and Tian \(2019\)](#).

predictions about the mapping between the quality of inventors’ ideas and their optimal choices. Section 4 calibrates the model. Section 5 simulates the model to evaluate the degree to which technology waves can account for changes in growth rates and market concentration through the idea allocation channel. Section 6 concludes.

2 Empirical Patterns

This section exhibits empirical patterns of the technology waves, idea allocation, and market concentration. A description of the data is provided in Appendix A.

2.1 Technology Waves

Technology waves capture the aggregate breakthrough intensity of new technologies over time. At the peak of a technology wave, there is a surge of mass innovation driven by high-quality, novel, and influential ideas. At the trough, most developments are incremental improvements upon existing ones, while next-generation technologies may still be in an early stage. Importantly, because this dynamic captures the combined novelty and influence of all technologies in each year, it is distinct from the diffusion process of a technology from emergence to full economy-wide dispersion, which usually takes decades, as documented by Atkeson and Kehoe (2007) and Kalyani et al. (2025).

2.1.1 Measurement

We measure the aggregate intensity of technological breakthroughs using three methods. The first is the “RETech” index developed by Bowen III, Frésard and Hoberg (2023), an aggregate-level measure that leverages patent-text vocabulary to quantify whether overall innovation is concentrated in rapidly evolving versus stable technological areas.² The second relies on the patent-importance metric developed by Kelly et al. (2021), which evaluates importance at the individual patent level by scaling a patent’s forward similarity to future patents by its backward similarity to prior art. We calculate the aggregate breakthrough intensity by taking the average of this metric across all patents granted in each year. However, the measure is truncated at 2005. Therefore, we construct a third, citation-based Technology Wave Index (TWI), extending the conceptual logic of Kelly et al. (2021) beyond 2005. TWI captures the same ratio of forward-looking impact to

²A related measure by Kalyani (2024) uses new patent phrases to capture creativity.

backward-looking reliance using patent citations. Specifically, for each year t , we define:

$$TWI_t = \frac{\sum_{i \in I_t} \sum_{s=0}^5 \text{Forward Citations}_{i,t+s}}{\sum_{i \in I_t} \sum_{s=0}^5 \text{Forward Citations}_{i,t+s} + \sum_{i \in I_t} \sum_{s=0}^5 \text{Backward Citations}_{i,t-s}}, \quad (1)$$

where I_t is the set of new patents granted in year t . The numerator sums the forward citations (citations received from others) that each new patent accumulates over the subsequent five years. The denominator scales this by the sum of these forward citations plus the backward citations (citations made to prior patents) to patents granted within the previous five years. The fixed five-year windows ensure that cohorts are compared on common ground, mitigating right-censoring issues. To smooth short-term volatility, we apply a three-year moving average:

$$TWI_avg_t = \frac{1}{3} \sum_{h=-1}^1 TWI_{t+h}.^3 \quad (2)$$

Panel A of Figure 1 plots our citation-based TWI_avg_t alongside the aggregate text-based series from Kelly et al. (2021) and Bowen III, Frésard and Hoberg (2023). They exhibit remarkably similar trends and fluctuations with largely synchronized peaks and troughs. This co-movement confirms that the identified technology waves reflect deep structural shifts in the innovation frontier rather than measurement artifacts. Across all measures, major technological breakthroughs occur in the mid-1980s, the mid-1990s, and the early 2010s, while the periods around 1990 and the mid-2000s represent phases of predominantly incremental innovation.⁴

One concern with patent-based measures is that they may reflect mechanical factors, such as constraints on backward citations or changes in aggregate patenting volume. To address this concern, Panel B of Figure 1 plots average forward and backward citations per patent and the total number of granted patents over time. The technology-wave measures primarily track forward citations, while backward citations exhibit a similar pattern with a lag, consistent with the legacy of earlier breakthroughs. Patent counts do not display the same fluctuations, suggesting that movements in the wave indices reflect distinct surges in relative patent quality and novelty rather than changes in patent volume. Field-level indices are reported in Figure 10 in Appendix B.1; they exhibit broad comovement alongside substantial heterogeneity across technological fields.

³The smoothing technique does not alter the underlying cyclical patterns, as shown in Appendix B.5.

⁴Although our primary analysis focuses on the US economy, we compute the citation-based TWI for six high-patenting European countries using PATSTAT. As shown in Figure 12 in Appendix B.4, the broad secular decline in breakthrough intensity from the 1980s through the 2010s is also present across these major European patenting economies.

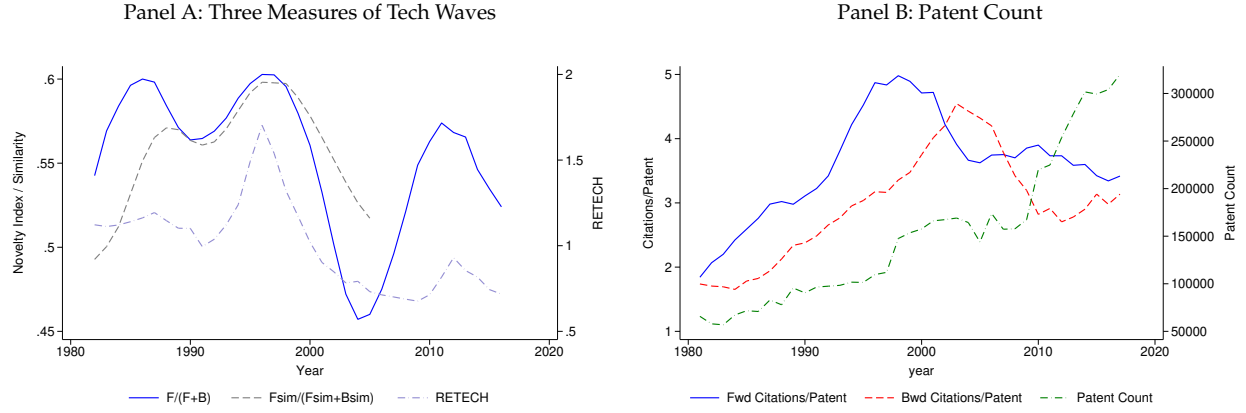


Figure 1: Technology Waves

Notes: Panel A illustrates three measures of technology waves. The blue solid curve is the citation-based measure defined in Equation 1, while the gray dashed curve is based on the text-similarity metric in Kelly et al. (2021); both use the left y-axis. The purple dash-dot curve shows the “RETech” index from Bowen III, Frésard and Hoberg (2023), which assesses novelty via the prevalence of growing vocabularies in patent descriptions, plotted on the right y-axis. Panel B plots the annual average numbers of forward and backward citations per patent and the total number of patents granted each year.

2.1.2 Contributors to the Tech Waves

Which classes of technology contributed to the three wave peaks? Who were the major applicants for breakthrough patents—incumbents, startups, or public institutions?

We address these questions in two steps (with full details in Appendix B.2 and B.3). First, we decompose TWI_t into contributions by three-digit IPC classes. The results highlight shifting technological frontiers: *Medical or Veterinary Science and Hygiene* is the largest contributor to the first peak (1985–1987), while *Computing; Calculating or Counting* becomes the dominant driver in both the second (1995–1997) and third (2010–2012) peaks, alongside *Electric Communication Technique*.

Second, we use the “historically significant patents” of Kelly et al. (2021) to identify breakthrough innovators. Breakthroughs originate broadly: 41% from incumbents, 26% from startups, and 20% from universities. These diverse origins motivate our focus on how technological opportunities shape the subsequent allocation of ideas across startups and incumbent firms.

2.2 Allocation of Ideas

Using Census and USPTO patent data, we examine how technology waves relate to idea allocation between startups and incumbents and to the matching of idea quality with incumbent firm size.

2.2.1 Entrants or Incumbent Firms

Though affiliations at idea inception are not directly available, they can be inferred by observing the firm age at the time of patent issuance, from the combined USPTO and LBD data. Specifically, when a patent is granted to a firm aged zero to five years, it indicates that the initial idea was developed in a startup five years earlier. In contrast, when a patent is granted to a firm older than five years, it implies that the idea was developed internally by an incumbent firm. The five-year cutoff reflects average USPTO application-to-grant lags (about 2–3 years, as documented by the USPTO) and a comparable research completion period. Under these assumptions, the ratio of ideas in new firms to ideas absorbed by incumbents is measured as:

$$\text{New-to-Incumbent Ratio}_t = \frac{\sum_{i \in I_{t+5}} \text{Granted in Firm}(\text{Age} \leq 5)_{i,t+5}}{\sum_{i \in I_{t+5}} \text{Granted in Firm}(\text{Age} > 5)_{i,t+5}}, \quad (3)$$

where I_{t+5} denotes the set of patents granted five years after time t , and the indicators “Granted in Firm(Age ≤ 5)” and “Granted in Firm(Age > 5)” equal one if patent i is granted to a firm aged five years or less, or older than five years, respectively.

As with the Tech Wave Index, a three-year moving average is applied to smooth the time series, and the resulting trend is shown in Figure 2.⁵ The Tech Wave Index is also plotted for comparison. A declining trend is observed in both series over time. In addition, the New-to-Incumbent Ratio exhibits pronounced cyclicity, with its peaks and troughs closely aligned with those of the technology waves. The cross-correlation between the detrended New-to-Incumbent Ratio (x_t) and the detrended Tech Wave Index (y_{t+k}) is computed across different time lags and reported in Table 11 in Appendix B.7. The maximum absolute correlation, 0.612, occurs at $k = 0$, indicating that the two series move in tandem. This positive relationship is further corroborated by a more granular instrumental-variable analysis. Section 5.4.1 discusses the regression evidence together with the results of the quantitative simulations.

The positive correlation between technology waves and the New-to-Incumbent Ratio is evident across most major technological fields, including human necessities, performing operations, chemistry, textiles, fixed constructions, physics, and electricity. Graphs by field illustrating this relationship are presented in Figure 14 in Appendix B.6.

⁵Figure 13 in Appendix B.5 presents the unsmoothed series for both the New-to-Incumbent Ratio and the Tech Wave Index, which display a similar correlation structure to their smoothed counterparts.

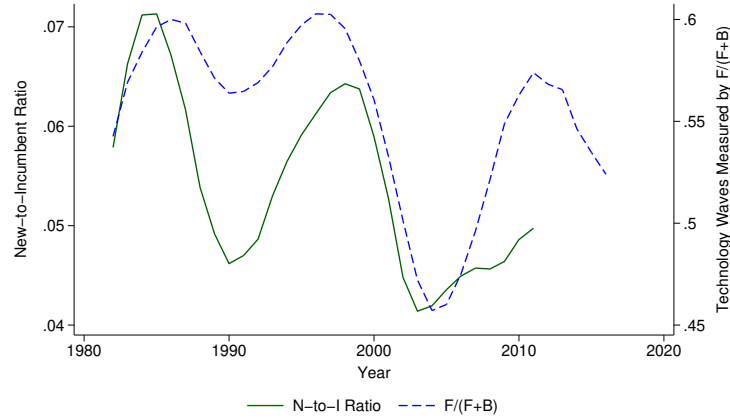


Figure 2: Technology Waves and Idea Allocation

Notes: This figure shows the technology waves and the idea allocation between new and incumbent firms over time. The blue dashed curve, based on the methodology defined in this paper, calculates the relative ratio of forward citations to the sum of forward and backward citations. The green solid curve displays the “New-to-Incumbent Ratio” defined in the paper, capturing where new ideas contribute their value. The two curves have different y-axes, which are shown on the left and right, respectively.

Sources: Longitudinal Business Database (LBD) and USPTO patent and citation data.

2.2.2 Size of Incumbent Firms

Beyond the division between startups and incumbents, we examine how idea quality varies with firm size within the incumbent group defined in Section 2.2.1. Idea quality is proxied by the number of forward citations each patent receives within five years of issuance, and firm size is measured by employment. We pool patents issued to both new and incumbent firms across years and compute citation quartiles to group patents by relative citation counts. Within each quartile, average assignee firm size (LBD employment) is computed, the lowest-quartile mean is normalized to one, and the resulting relative sizes are plotted in Figure 3.

The figure reveals a pattern of positive assortative matching between idea quality and firm size: higher-quality ideas are more likely to be matched with larger firms. However, we do not find this assortative matching pattern when calculating the average assignee size across citation quartiles for patents granted to firms younger than five years.

2.3 Patent Value and Equity Incentives over Technology Waves

The preceding evidence describes how innovation is allocated between startups and incumbents and across incumbent firm sizes. We now examine potential forces behind these patterns using patent-level evidence on economic value and firm-level evidence on equity compensation. Larger firms may enhance an idea’s value through greater

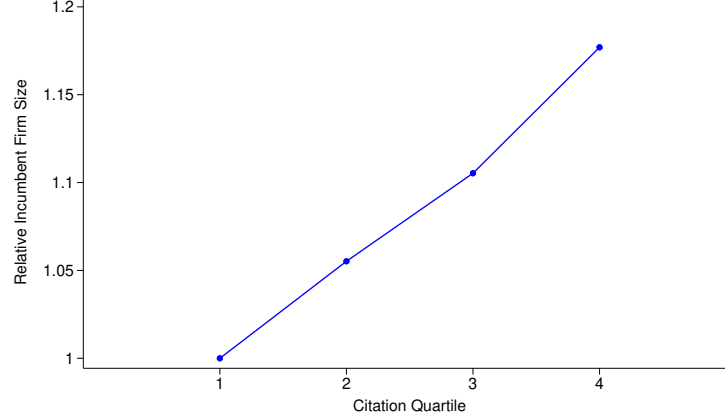


Figure 3: Mapping between Patent Citations and Incumbent Firm Size

Notes: This figure shows the mapping between inventors' idea quality and firm size if inventors opt to develop their ideas in incumbent firms. The idea quality is measured by the number of patent citations and is classified into four quartiles. The firm size is measured by the number of employees. The average employment of firms corresponding to the first citation quartile is normalized to be one.

Sources: Longitudinal Business Database (LBD) and USPTO patent and citation data.

commercialization capacity and complementary assets, while technological integration and incentive provision may limit these advantages. The analysis provides suggestive evidence on these forces and how they vary with technology waves and firm size.

We begin with the patent-level analysis, using an extended version of the data from [Kogan et al. \(2017\)](#). These data use the stock market's response to patent news to estimate the private economic value of patents granted to publicly listed U.S. firms. We estimate

$$\begin{aligned} \ln(\text{Economic value}_{ijt}) = & b \ln(\text{Firm size}_{j,t-1}) + \iota \ln(1 + \text{Citations}_{it}) \\ & + \phi \text{TWI}_{kt} \times \ln(1 + \text{Citations}_{it}) + \theta_{ct} + \epsilon_{ijt}, \end{aligned} \quad (4)$$

where i , j , c , and t index patents, firms, four-digit IPC classes, and years. The index k denotes either the aggregate economy or IPC class c . Firm size is measured by lagged sales, limiting the possibility that the market response to the current patent mechanically raises measured firm size. Patent quality is measured by forward citations received within five years. All specifications absorb IPC-by-year fixed effects. The specifications that omit firm size additionally absorb firm-by-year fixed effects, thereby comparing patents of different scientific quality issued by the same firm in the same year. A positive b would be consistent with larger firms providing synergies that raise the private value generated by a given idea. A negative ϕ would indicate that the conversion of scientific quality into private value weakens during stronger technology waves, a pattern potentially reflecting greater adoption frictions for frontier-shifting ideas.

We then turn to the firm-level analysis. One possible counterweight to the positive role of firm size is that equity-based compensation may provide weaker innovation incentives in larger firms (Holmstrom (1989); Lerner and Wulf (2007)). The data do not identify the contractual equity share received by individual inventors, so we use the following firm-level measure of stock compensation per employee as a proxy for the strength of equity-based incentives:

$$A_{jt} = \ln \left(1 + \frac{\text{stock compensation}_{jt}}{\text{employment}_{jt}} \right).$$

We first ask whether this incentive proxy varies systematically with firm size:

$$A_{jt} = \alpha + \gamma \ln(\text{sales}_{jt}) + \rho Q_{j,t-1} + \theta_{s(j)t} + \mu_j + u_{jt}, \quad (5)$$

where $Q_{j,t-1}$ is the logarithm of one plus the average forward citations of patents issued by firm j during years $t - 3$ through $t - 1$, and $\theta_{s(j)t}$ denotes SIC-by-year fixed effects. Firm fixed effects μ_j are included in the within-firm specifications. We next ask whether stronger predetermined equity-based incentives predict subsequent innovation value:

$$\begin{aligned} \ln \left(1 + \sum_i \xi_{ijt} \right) = & \alpha + \beta A_{j,t-1} + \delta Q_{j,t-1} + \eta \ln(\text{sales}_{j,t-1}) \\ & + \kappa A_{j,t-1} Q_{j,t-1} + \psi \ln(\text{sales}_{j,t-1}) Q_{j,t-1} + \theta_{s(j)t} + \mu_j + \varepsilon_{jt}, \end{aligned} \quad (6)$$

where $\sum_i \xi_{ijt}$ is the total Kogan et al. (2017) patent value produced by firm j in year t . Specifications without interactions set $\kappa = \psi = 0$. The interactions allow the incentive and size channels to vary with the quality of the firm's recent innovative opportunities. The firm-level sample covers 2006–2016 and includes firms that issued at least one patent during the preceding three years. Standard errors are clustered by firm.

Table 1 combines the patent- and firm-level evidence. Columns (1)–(4) report the patent-level analysis. Column (1) uses lagged sales to measure firm size, column (2) introduces firm-by-year fixed effects, and columns (3)–(4) interact patent quality with the aggregate and IPC-level Tech Wave Indexes. Columns (5)–(10) report the firm-level analysis, examining whether firm size predicts equity-based compensation and whether lagged compensation predicts total patent value, both across and within firms over time.

The patent-level estimates suggest a positive role for firm size. In column (1), the coefficient on lagged sales is 0.314 and statistically significant. This association is consistent with larger firms providing synergies that raise the private value generated by a given idea. The interaction between patent quality and the aggregate Tech Wave Index

Table 1: Firm Size, Equity Incentives, and Innovation Value

	Patent-level evidence			Firm-level incentive mechanism					
	(1)	(2)	(3)	(4)	Equity (5)	Total patent value (6)	Equity (8)	Total patent value (9)	Total patent value (10)
Lagged firm size	0.314*** (0.030)					0.849*** (0.025)		0.274*** (0.033)	0.211*** (0.038)
Patent quality	0.094*** (0.018)	0.004*** (0.001)	0.028** (0.012)	0.013* (0.006)					
Patent quality $\times$ TWI (year)			-0.045** (0.022)						
Patent quality $\times$ TWI (IPC-year)				-0.016 (0.011)					
Firm size					-0.132*** (0.015)		-0.050** (0.020)		
Lagged equity incentive						0.289*** (0.023)	0.189*** (0.031)	0.039*** (0.014)	-0.010 (0.024)
Prior patent quality					0.364*** (0.033)	0.634*** (0.044)	-2.813*** (0.450)	0.239*** (0.039)	-0.884*** (0.306)
Lagged equity incentive $\times$ prior quality						0.068*** (0.017)		0.032*** (0.012)	
Lagged firm size $\times$ prior quality						0.151*** (0.022)		0.045*** (0.015)	
Observations	1,186,261	1,198,027	1,198,027	1,198,027	12,988	12,650	12,650	12,326	12,326
R^2	0.437	0.883	0.883	0.883	0.545	0.627	0.638	0.901	0.901
IPC $\times$ year FE	Yes	Yes	Yes	Yes	No	No	No	No	No
Firm $\times$ year FE	No	Yes	Yes	Yes	No	No	No	No	No
SIC $\times$ year FE	No	No	No	No	Yes	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	No	No	Yes	Yes	Yes

Notes: Columns (1)–(4) report patent-level regressions. Column (1) uses lagged sales; columns (2)–(4) absorb firm-by-year fixed effects. Patent quality is $\ln(1 + \text{five-year forward citations})$. Standard errors in these columns are clustered by four-digit IPC class and year. Columns (5)–(10) report firm-year regressions for previous patenters in 2006–2016, including current zero-patent years. Equity incentive is $\ln(1 + \text{stock compensation per employee})$; prior patent quality is the logarithm of one plus average forward citations on the firm's patents in years $t - 3$ through $t - 1$; and total patent value is $\ln(1 + \sum_i \xi_{it})$. Standard errors in these columns are clustered by firm. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

is negative in column (3), suggesting that the conversion of scientific quality into private value weakens during stronger technology waves. This pattern is suggestive of greater adoption frictions when technologies are advancing more rapidly. The corresponding IPC-by-year interaction in column (4) is also negative in the current specification.

The firm-level estimates suggest a possible incentive-related counterweight. Larger firms provide less stock compensation per employee: the coefficient on firm size is -0.132 in column (5) and remains negative at -0.050 after firm fixed effects are included in column (8). Stronger lagged equity-based compensation, in turn, predicts greater total patent value. The coefficient is 0.289 in column (6) and remains positive and statistically significant at 0.039 with firm fixed effects in column (9). Thus, the association is not explained solely by permanent differences across firms, although it should not be interpreted as causal. Columns (7) and (10) further suggest that equity-based incentives may be more important when recent innovative opportunities are of higher quality. The interaction coefficient is positive and significant with and without firm fixed effects.

Taken together, these results point to three potential forces shaping idea allocation: synergies with firm size, adoption frictions for frontier-shifting ideas, and weaker innovation incentives in larger firms.

2.4 Market Concentration

Finally, this section examines the relationship between technology waves and market concentration. Our primary measure of market concentration is the Herfindahl-Hirschman Index (HHI), constructed using two datasets: the Census Bureau's Longitudinal Business Database (LBD) for all U.S. employers, and Compustat Fundamentals Annual for publicly listed U.S.-headquartered industrial firms. We compute annual industry-level HHIs by summing squared firm market shares—defined by employment or payroll at the 6-digit NAICS level in the LBD, and by sales at the 4-digit SIC level in Compustat. We then aggregate these into an economy-wide HHI using industry employment or sales weights, respectively. Finally, we apply a three-year moving average to smooth the series.⁶

Panel A of Figure 4 displays the annual Herfindahl-Hirschman Index (HHI) for firm employment, payroll, and sales, all of which exhibit similar trends and fluctuations, except for the last ten years. The pairwise correlations among these measures are high: 0.96 between employment and payroll, 0.73 between employment and sales, and 0.66 between payroll and sales. Panel B illustrates the relation between market concentration, measured by the employment-based HHI, and the technology waves measured by TWI.

⁶Figure 15 in Appendix B.5 plots the unsmoothed LBD employment HHI and Tech Wave Index, which exhibit a similar correlation.

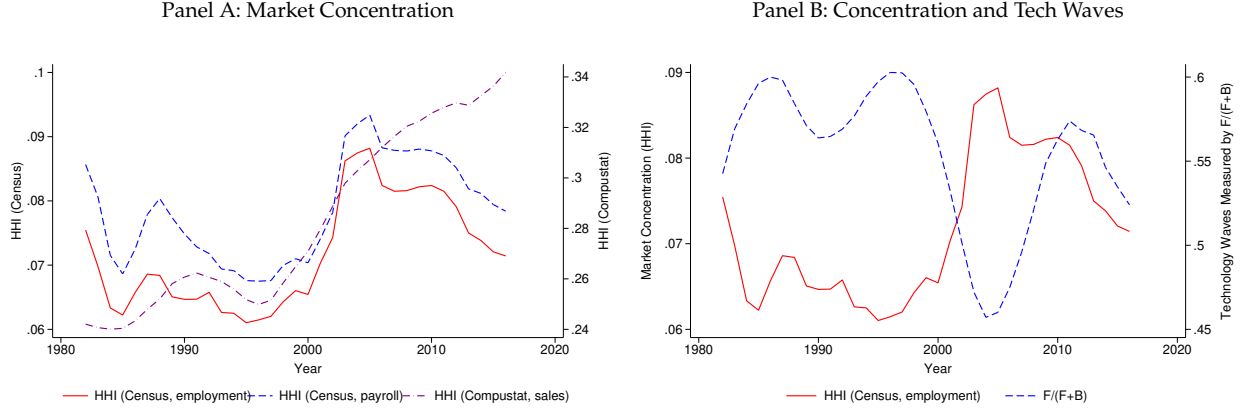


Figure 4: Technology Waves and Market Concentration

Notes: Panel A displays the annual HHI for employment (red solid curve), payroll (blue dashed curve), and sales (purple dash-dot curve). The first two are derived from the LBD dataset from the Census, while the last one is based on Compustat. Panel B illustrates the technology waves alongside the trend of market concentration over time. The red solid curve shows the HHI for employment. The blue dashed curve, following the methodology defined in this paper, represents the relative ratio of forward citations to the sum of forward and backward citations. The scale of y-axes are shown on the left and right, respectively. *Sources:* Longitudinal Business Database (LBD), Compustat Fundamental Annuals, and USPTO patent data.

The two series are negatively correlated. The technology waves exhibit a downward linear trend, while the HHI shows an upward trend. The cross-correlations between the detrended HHI (x_t) and the detrended technology waves (y_{t+k}) are -0.736 , -0.775 , and -0.720 respectively when k equals 0 , -1 , -2 . This suggests that changes in market concentration closely follow technology waves with a lag.⁷

To assess the robustness of market concentration patterns, the share of sales by top firms is calculated using the cleaned data series from [Kwon, Ma and Zimmermann \(2023\)](#), which is based on IRS data covering the entire population of U.S. corporations. Figure 16 in Appendix B.9 shows that the top sales shares exhibit similar upward trends and cyclical patterns to the HHI.⁸ A more granular analysis using regressions with instrumental variables confirms the negative relationship between the technological waves and the market concentration. These findings, together with the quantitative simulation results, are presented in Section 5.4.1.

⁷Cross-correlations for $k \in \{-3, -2, -1, 0, 1, 2, 3\}$ are reported in Table 11 in Appendix B.7. The strongest correlation, in absolute value, occurs at $k = -1$.

⁸The aggregate negative correlation between technology waves and market concentration is also evident in Europe, as shown by the declining trend of the Tech Wave Index in Appendix B.4 and the increasing market concentration across European countries, as measured by the Herfindahl-Hirschman Index (HHI) and top sales share in recent studies (e.g., [Bighelli et al. \(2023\)](#) and [Ma, Zhang and Zimmermann \(2024\)](#)).

3 Model

To formalize how technology waves drive market concentration, we develop a general equilibrium model with two groups of individuals—households and inventors—and two types of firms: intermediate goods producers and final goods producers. In each period, the inventors draw ideas of heterogeneous quality. They decide either to establish new intermediate-goods firms (Romer-type innovation) or to join incumbent firms of chosen size (quality-ladder innovation). The economy is subject to aggregate shocks that vary the scale of the idea quality distribution. The shocks shift the relative payoff to inventors across innovation types. Inventors' choices depend jointly on the aggregate shock and the quality of their individual ideas.

3.1 Preferences

There is a long-lived representative household in the economy. She works in the production sector, supplies one unit of labor to firms inelastically, and consumes final goods. She also owns all the firms in the economy. The household's utility function is

$$U_H = \int_0^\infty e^{-\rho t} \log(C_H(t)) dt, \quad (7)$$

where $\rho > 0$ is the discount rate and $C_H(t)$ is the consumption of the household.

Inventors are the ones who work in the R&D sector. In each period, there is a continuum of inventors of measure one. An inventor, who lives for dt time periods, dedicates effort $e_I(t)$ to create innovations within either an incumbent firm or a new business and simultaneously consumes. Inventors are risk-averse and have a mean-variance utility flow:

$$U_I(c_I(t), e_I(t)) = \mathbb{E}(c_I(t)) - A \frac{\text{var}(c_I(t))}{V(t|Q(t))} - R(e_I(t)) V(t|Q(t)), \quad (8)$$

where $c_I(t)$ is the consumption, $e_I(t)$ is the effort level, and $R(e_I) V(t|Q(t))$ is the associated cost. $Q(t)$ (defined below) is the aggregate quality in the economy at time t , and $V(t|Q(t))$ is the value of a firm whose size equals the aggregate quality. The variance and cost are normalized by $V(t|Q(t))$ to keep the problem stable over time. Denote the inventors' aggregate consumption by $C_I(t)$, i.e., $C_I(t) = \int_0^1 c_{Ii}(t) di$.

3.2 Technology

The economy features two types of firms: intermediate goods producers and final goods producers, as in [Akcigit and Kerr \(2018\)](#). Both types of firms are owned by the household. The former hires inventors to create innovations, and produces intermediate goods. The latter assembles intermediate goods and produces final goods.

The final good producers produce final goods using a continuum of intermediate goods $j \in [0, N_F(t)]$:

$$Y(t) = \frac{1}{1-\beta} \int_0^{N_F(t)} q_j^\beta(t) y_j^{1-\beta}(t) dj. \quad (9)$$

In this function, $q_j(t)$ is the quality of the intermediate good j , and $y_j(t)$ is its quantity. $N_F(t)$ is the total mass of intermediate goods producers. We normalize the price of the final good to be one in every period. The final good producers are perfectly competitive, taking the input prices as given.

The intermediate goods producers are a continuum of risk-neutral firms. Each firm produces one type of good, with a linear technology using only labor:

$$y_j(t) = Q(t) l_j(t), \quad (10)$$

where $l_j(t)$ is the labor input; $Q(t) = \int_0^{N_F(t)} q_j(t) dj$ is the aggregate quality level of the economy. It implies that improvements in both q_j (intensive margin) and N_F (extensive margin) have positive externalities ([Romer, 1986](#); [Aghion and Howitt, 1990](#)). The cost is linear in wage $w(t)$, which intermediate firms take as given. The labor market satisfies the constraint:

$$\int_0^{N_F(t)} l_j(t) dj \leq 1. \quad (11)$$

3.3 Innovation

Innovation takes two forms: creating new intermediate-good varieties (*Romer-type*) or improving the quality of existing lines (*Quality-ladder*). *Romer-type* innovation is led by an inventor pairing with an outside, risk-neutral partner. *Quality-ladder* innovation occurs within incumbent firms, led either by an inventor partnering with the incumbent or by the incumbent itself after acquiring the idea. Because inventor effort is unobservable, a principal-agent problem featuring moral hazard arises whenever the inventor leads the project ([Holmstrom, 1989](#)). We assume an exogenous share $1 - h$ of ideas is best implemented entirely by incumbents; these are sold to incumbents of random size and follow a standard quality-ladder process. For the remaining share h , the inventors choose to lead innovations in either startups or incumbents of an optimal size based on

equilibrium contracts. This organizational choice trades off adoption frictions, incumbent synergies, and incentive alignment. Table 2 summarizes the features of each innovation method, which are detailed in the following sections.

Table 2: Features of Different Innovation Methods

Innovation Type	Leader	Partner	Adoption Frictions	Synergy	Incentive Problems	Share
Romer	Inventor	Partner	No	No	Yes	h
Quality-ladder	Inventor	Incumbent	Yes	Yes	Yes	
Quality-ladder	Incumbent	None	Yes	Yes	No	$1 - h$

3.3.1 Inventor-led R&D

The key to R&D management is finding the optimal incentive structure to motivate unobservable effort (Aghion and Tirole, 1994; Merges, 1999). Consistent with empirical evidence demonstrating that compensation structures directly drive both inventor effort and the allocation of innovation (Lerner and Wulf, 2007; Giarratana, Mariani and Weller, 2018), we explicitly model the principal-agent problem between inventors and firms. We assume all partners employ a widely adopted contract structure that combines a fixed wage for risk-sharing with an equity component for incentive alignment (Onishi, 2013).⁹ Formally, the inventors endogenously choose an effort level, e_I , which governs the innovation arrival rate according to a Poisson process:

$$\lambda(e_I) = \lambda_0 e_I. \quad (12)$$

Exerting effort is costly for inventors. The instantaneous cost of effort e_I is given by $R(e_I)V(t|Q(t))$, where $R(e_I) = \frac{1}{1+\delta}e_I^{\delta+1}$, an increasing and convex function of effort.¹⁰

Each inventor is endowed with a single innovative idea of quality z_0 , drawn randomly from a distribution, $\Psi(z_0)$. If the inventor leads innovation in a startup, there is full pass-through from idea quality to realized innovation quality, $zQ(t)$, which is drawn

⁹Because the inventor's effort level, e_I , is unobservable and unverifiable, contracts cannot be directly contingent on it. Consequently, firm equity is used to align the inventor's returns with the firm's.

¹⁰The innovation production function and cost specifications follow the growth-theory literature (Romer, 1990; Klette and Kortum, 2004; Akcigit and Kerr, 2018). In our calibration, we set $\delta = 1$, as is standard. We model inventors' R&D costs as a dis-utility. Equivalently, they can be represented as a resource cost that converts one-to-one into final goods. This modeling choice has no material effect on our results.

from a uniform distribution,

$$\text{Uniform}((1 - \phi)z_0Q(t), (1 + \phi)z_0Q(t)). \quad (13)$$

On average, higher-quality ideas lead to higher-quality innovations, but the parameter $\phi \in (0, 1)$ introduces randomness into this mapping by capturing variability in how idea quality translates into the realized outcome.

If the inventor with idea quality z_0 leads co-innovation with an incumbent firm of quality q , the quality improvement, $x(z_0, \tilde{q}(t))Q(t)$, reflects characteristics of both the inventor and the firm. We model it as a stochastic draw from a uniform distribution:

$$\text{Uniform}((1 - \phi)x_0(z_0, \tilde{q}(t))Q(t), (1 + \phi)x_0(z_0, \tilde{q}(t))Q(t)), \quad (14)$$

where $\tilde{q}(t)$ denotes the incumbent's relative quality, defined as $\tilde{q}(t) = \frac{q}{Q(t)}$. For convenience in the subsequent analysis, we adopt the normalization $\tilde{m} = \frac{m}{Q(t)}$ for variable m when necessary. The effective idea quality, $x_0(z_0, \tilde{q}(t))$, depends positively on both the inventor's original idea quality and the relative firm size, i.e., $\frac{\partial x_0(z_0, \tilde{q}(t))}{\partial z_0} > 0$ and $\frac{\partial x_0(z_0, \tilde{q}(t))}{\partial \tilde{q}(t)} > 0$. The function $x_0(z_0, \tilde{q}(t))$ takes the form:

$$x_0(z_0, \tilde{q}(t)) = \chi(\tilde{q}(t))\gamma(z_0) = \left(\frac{\tilde{q}(t)}{q_0}\right)^b (B^\eta + z_0^\eta)^{\frac{1}{\eta}}, \quad \eta < 1, b > 0.$$

The first term, $\left(\frac{\tilde{q}(t)}{q_0}\right)^b$, capturing incumbents' synergy benefits, rises with the firm's relative quality. The second term, $(B^\eta + z_0^\eta)^{\frac{1}{\eta}}$ with $\eta < 1$, is a constant-elasticity-of-substitution (CES) aggregator that captures the interaction between the inventor's idea and the incumbent's existing technology base B in the co-invention process. The CES structure with imperfect substitution between the new idea z_0 and the technological base, B , implies that the new idea cannot fully substitute for existing product lines within incumbent firms. So, idea quality does not fully pass through to realized economic value, reflecting frictions in adopting new technologies. This reduced-form approach to modeling compatibility costs follows the spirit of [Jovanovic and Rousseau \(2014\)](#). This specification separates the synergy from the co-invention process, assuming that all firms share the same technology base. An alternative functional form that embeds firm size and the technology base together in the CES aggregator is discussed in [Appendix C.11](#).

Joining Incumbent Firms: Incumbent intermediate firms compete à la Bertrand to attract inventors by offering compensation packages consisting of wage rate $T_I V(t|Q(t))$

and equity a . The firm's quality, and equivalently its size, evolves stochastically:

$$dq = -q dJ_{\text{exit}} + x(z_0, \tilde{q}(t))Q(t)dJ_{R\&D}(e_I), \quad (15)$$

where J_{exit} and $dJ_{R\&D}(e_I)$ are two jump processes with Poisson arrival rates at τ and $\lambda(e_I) = \lambda_0 e_I$, respectively. The first jump process represents the exogenous exit, while the second one is related to the in-house innovation. The magnitude of the jump is $x(z_0, \tilde{q}(t))Q(t)$, the stochastic value of an innovation produced by inventor z_0 in a firm with a normalized size $\tilde{q}(t)$ when the aggregate technology level is $Q(t)$.¹¹

The inventor's compensation $c_I(a, T_I, q, z_0, Q(t))$ is:

$$c_I(a, T_I, q, z_0, Q(t))dt = T_I V(t|Q(t))dt + adV(j, t|q, z_0), \quad (16)$$

where $dV(j, t|q, z_0)$ is the firm value change from t to $t + dt$ when hiring an inventor of quality z_0 . Following a principal-agent framework, the compensation package includes, in addition to a fixed wage, an opportunity for the inventor to purchase a fraction a of equity at the fair price. At the end of each period of length dt , the inventor sells this equity back to the firm to finance consumption. In doing so, the inventor bears part of the risk from exit shocks and innovation shocks. A firm j with quality q solves the following problem when an inventor with idea quality z_0 leads the innovation process:

$$\begin{aligned} \Omega_I(z_0, q, t) &= \max_{a, T_I} (1 - a) \lambda_0 e_I \left[\int V(j, t|q + x(z_0, \tilde{q}(t))Q(t))dx - V(j, t|q) \right] - T_I V(t|Q(t)) \\ \text{s.t. } e_I &= \arg \max \{U_I(c_I(a, T_I, q, z_0, Q(t)), e_I)\} \\ U_I(c_I(a, T_I, q, z_0, Q(t)), e_I) &\geq \bar{u}(z_0, t) \\ (1 - a) \lambda_0 e_I \left[\int V(j, t|q + x(z_0, \tilde{q}(t))Q(t))dx - V(j, t|q) \right] &- T_I V(t|Q(t)) \geq 0 \end{aligned} \quad (17)$$

where $\Omega_I(z_0, q, t)$ denotes the expected payoff. The instant change rate in a firm's expected payoff from hiring the inventor equals the increase in firm value from the innovation accruing to the original shareholders (all shareholders other than the inventor), net of the wage and equity granted to the inventor. Here, $V(j, t|q)$ denotes the firm's value without R&D.

The first constraint is the inventor's incentive-compatibility condition: given the contract a, T_I , she chooses effort e_I to maximize expected utility $U_I(c_I(a, T_I, q, z_0, Q(t)), e_I)$. The second is the inventor's participation constraint,

¹¹The parameter b in $x_0(z_0, \tilde{q}(t))$, estimated from the data, governs how strongly the economic value depends on firm quality, $q(t)$, relative to the aggregate quality, $Q(t)$.

requiring utility of at least $\bar{u}(z_0, t)$, where the outside option is endogenously pinned down by Bertrand competition. The third is the firm's participation constraint, ensuring the firm is not worse off from hiring the inventor.

The inventor's utility function can be expressed as:

$$U_I(c_I(a, T_I, q, z_0, Q(t)), e_I) = \mathbb{E}(c_I(a, T_I, q, z_0, Q(t))) - A \frac{\text{Var}(c_I(a, T_I, q, z_0, Q(t)))}{V(t|Q(t))} - R(e_I) V(t|Q(t)). \quad (18)$$

Starting up a New Business: If an inventor starts a new business with an outside partner, the partner solves an optimization problem analogous to that of an incumbent firm (Equation (17)). The key differences are that there is neither synergy nor adoption friction, so $q = 0$ and $x_0 = z_0$. We assume the partner's outside option value is zero, so its participation constraint also requires nonnegative net payoff. The inventor then chooses her effort level to maximize utility.¹²

3.3.2 Firm-led R&D

If an incumbent firm leads the innovation, it purchases an idea from an inventor and selects its own effort level e_F . The innovation's success is then governed by an instantaneous Poisson arrival rate:

$$\lambda(e_F) = \lambda_0 e_F. \quad (19)$$

Exerting effort is also costly for firms. The instantaneous cost of effort e_F is given by $R(e_F)N_F(t)V(t|q(t))$, where $R(e_F) = \frac{1}{1+\delta}e_F^{\delta+1}$.¹³ Both the cost function $R(\cdot)$ and the arrival rate $\lambda(\cdot)$ take the same functional forms as in the inventor-led innovation case. The total R&D cost borne by firms is:

$$R_F(t) = \int_0^{N_F(t)} R(e_F(t, q(j)))N_F(t)V(t|q(t))dj. \quad (20)$$

Since firms bear both the costs and the benefits of innovation, they are fully incentivized to choose the effort level that maximizes their payoff. If the purchased idea has quality z_0 , a successful innovation raises the firm's quality from $q(t)$ to $q(t) + x_F N_F(t)q(t)$, where $x_F N_F(t)q(t)$ is drawn from a uniform distribution similar to (13):

¹²A detailed exposition of the startup problem is provided in Appendix C.1.

¹³The introduction of $N_F(t)V(t|q(t))$ makes the cost and benefit of firm-led R&D comparable with innovator-led R&D.

$$\text{Uniform}((1 - \phi)\kappa x_{F0}N_F(t)q(t), (1 + \phi)x_{F0}N_F(t)q(t)), \quad (21)$$

where $x_{F0} = \gamma(z_0) = (B^\eta + z_0^\eta)^{\frac{1}{\eta}}$. $\kappa > 0$ captures the scale effects of a firm-led innovation. As compensation, the firm makes a fixed transfer $T_F N_F(t) V(t|q(t))$ to the inventor. The firm maximizes the expected value of innovation by solving the following problem,

$$\begin{aligned} \Omega_F(z_0, q, t) = \max_{e_F, T_F} \lambda_0 e_F \left[\int V(j, t|q + \kappa x_{F0} N_F(t) q) dz - V(j, t|q) \right] \\ - \frac{1}{1 + \delta} e_F^{\delta+1} N_F(t) V(t|q(t)) - T_F(t) N_F(t) V(t|q(t)). \end{aligned} \quad (22)$$

This problem is similar to a standard quality-ladder model. As in the inventor-led case, we assume that the transfer T_F equals the firm's expected innovation value.¹⁴

$$\lambda_0 e_F \left[\int V(j, t|q + \kappa x_{F0} N_F(t) q) dz - V(j, t|q) \right] - \frac{1}{1 + \delta} e_F^{\delta+1} N_F(t) V(t|q(t)) = T_F N_F(t) V(t|q(t)) \quad (23)$$

3.4 Timeline

At birth, each inventor observes the initial quality z_0 of her idea. A share $1 - h$ of ideas is sold to an incumbent of random size. For the remaining share h , an outside partner—who also observes z_0 —offers a contract to jointly establish a new intermediate firm. At the same time, incumbent firms observe their corresponding $x_0(z_0, \tilde{q})$ and extend employment contracts to the inventor. After evaluating all offers, the inventor either joins her preferred incumbent with relative quality $\tilde{q}^*(z_0, t)$ or launches a startup with the outside partner.

The inventor's decision rule is:

$$u(z_0, t) = \max\{U_I(c_I(z_0, q^*(z_0, t), t), e_I(z_0, q^*(z_0, t), t)), U_I(c_I(z_0, 0, t), e_I(z_0, 0, t))\}. \quad (24)$$

3.5 Entry and Exit

Incumbent intermediate firms face an exogenous exit rate τ , which is independent of their size and is a risk unrelated to innovation.¹⁵ After a firm exits the market, the technology it owns is distributed to new entrants randomly. Entry of new intermediate firms occurs

¹⁴An alternative assumption is that the incumbent captures all surplus. The resulting expressions are of a similar functional form, since firm value remains linear in size.

¹⁵A modest negative correlation between exit rates and firm size does not affect the model's predictions.

through successful innovations by inventors launching startups with outside partners. The entry rate is denoted by $\lambda_I(t)$:

$$\lambda_I(t) = \int_{z_0 \in \{q^*(t)=0\}} h \lambda_0 e_I(z_0, 0, t) d\Psi(z_0). \quad (25)$$

Upon entry, an entrant draws a relative quality level $\tilde{q}_I$ from the distribution $f_{I,t}(\tilde{q}_I)$, where it satisfies

$$f_{I,t}(\tilde{q}_I) = \frac{\lambda_I(t)}{\tau N_F(t)} f_t(\tilde{q}), \tilde{q}_I = \frac{\tau N_F(t)}{\lambda_I(t)} \tilde{q}.$$

where $f_t(\tilde{q})$ is the incumbents' quality distribution. Therefore, the total quality flow of exiting incumbents is equal to the total quality flow of entrants,

$$\lambda_I(t) \int q_I f_{I,t}(q_I) dq_I = \lambda_I(t) \int \tilde{q}_I Q(t) f_{I,t}(\tilde{q}_I) d\tilde{q}_I = N_F(t) \int \tau \tilde{q} Q(t) f_t(\tilde{q}) d\tilde{q} = \tau Q(t). \quad (26)$$

The new firm incurs a cost equal to the value of the technology associated with its drawn quality, $V(t|q_I)$. Its quality is then updated by the inventor's innovation.

The law of motion for the mass of intermediate-goods firms is

$$\dot{N}_F(t) = \lambda_I(t) - \tau N_F(t). \quad (27)$$

3.6 Household's and Firms' Choices

The household maximization problem reveals the relationship between the growth rate, $g(t)$, and the time discount factor,

$$\frac{\dot{C}_H(t)}{C_H(t)} = r(t) - \rho. \quad (28)$$

The final good producer chooses $\{y_j(t)\}_j$ to maximize its profit using the technology described in Section 3.2, which yields the demand function faced by intermediate goods producers: $p_j(t) = q_j(t)^\beta y_j(t)^{-\beta}$. The intermediate good producers engage in monopolistic competition.¹⁶ Their FOC yields,

$$y_j(t) = q_j(t) \left(\frac{Q(t) (1 - \beta)}{w(t)} \right)^{\frac{1}{\beta}}, l_j(t) = \frac{y_j(t)}{Q(t)}, p_j(t) = \frac{w(t)}{Q(t) (1 - \beta)}. \quad (29)$$

¹⁶The profit maximization problem and the solution process of the final good and intermediate good producers are shown in Appendix C.2.

In each period, the labor market clearing condition pins down the wage:

$$w(t) = (1 - \beta) Q(t). \quad (30)$$

Thus, both the production output $y_j(t)$ and profit $\pi_j(t)$ are linear in quality,

$$y_j(t) = q_j(t), \pi_j(t) = \beta q_j(t), Y(t) = Q(t). \quad (31)$$

The HJB equation of an incumbent firm's value is

$$\begin{aligned} r(t)V(j, t|q) - \dot{V}(j, t|q) &= \pi_j(t) + \Omega_I(z_0, q, t) + \Omega_F(z_0, q, t) - \tau V(j, t|q) \\ &= \pi_j(t) - \tau V(j, t|q) \end{aligned} \quad (32)$$

Inventors are the scarce factor in the innovation process and they cannot substitute each other. We assume this competition takes the Bertrand form. Incumbents and outside partners simultaneously post contracts (a, T_I) , and each inventor selects the most attractive offer. Because firms are ex ante symmetric in their access to any given idea, Bertrand competition bids the transfer up until the marginal firm earns zero rent from the match, $\Omega_I(z_0, q, t) = \Omega_F(z_0, q, t) = 0$. Inventors therefore capture the entire surplus from innovation, and incumbent value reflects production profits,

$$r(t)V(j, t | q) - \dot{V}(j, t | q) = \pi_j(t) - \tau V(j, t | q),$$

which delivers the linear value function $V(j, t | q) = v(t)q$. This leads to the linearity that underlies the characterization in Proposition 1.¹⁷

Since $\pi_j(t) = \beta q_j(t)$, the value of an intermediate firm is linear in its quality, i.e.,

$$V(j, t|q) = v(t)q. \quad (33)$$

Solving the firm-led innovation problem yields the incumbent firm's effort, $e_F = (\lambda_0 \kappa x_{F0})^{\frac{1}{\delta}}$, which is invariant to time and firm size¹⁸. The arrival rate, $\lambda(e_F) = \lambda_0^{\frac{1+\delta}{\delta}} (\kappa x_{F0})^{\frac{1}{\delta}}$, is constant across firms. The transfer to the inventor, $T_F(t)N_F V(t|q(t)) = \frac{\delta}{1+\delta} (\lambda_0 \kappa x_{F0})^{\frac{1+\delta}{\delta}} v(t)N_F q(t)$, is linear in $v(t)$. Inventors' ideas are sold to firms randomly.

Solving the inventor-led innovation problem shows that the inventor's effort, e_I , is independent of time when (33) holds. The inventor's consumption, $c_I(a, T_I, q, z_0, Q(t))$, and utility, $U_I(c_I, e_I)$, are both linear in $v(t)$. The inventor's optimal firm choice is

¹⁷Appendix C.12 considers a case where firms and inventors share the surplus. It yields similar results.

¹⁸See Appendix C.3 for details.

invariant to time or $\nu(t)$ and depends only on parameters.¹⁹

3.7 Aggregate Variables

The final goods are consumed by the household and inventors, or invested into R&D by firms. The resource constraint of the economy is:

$$Y(t) = C_H(t) + C_I(t) + R_F(t), \quad (34)$$

Inventors' total consumption, $C_I(t)$, by applying the law of large numbers:

$$\begin{aligned} C_I(t) = & \nu(t)Q(t) \int_{z_0 \in \{z_0 | \tilde{q}^* > 0\}} (h\lambda_0 e_I(z_0, \tilde{q}^*) x_0(z_0, \tilde{q}^*) + (1-h)T_F) d\Psi(z_0) \\ & + \nu(t)Q(t) \int_{z_0 \in \{z_0 | \tilde{q}^* = 0\}} (h\lambda_0 e_I(z_0, \tilde{q}^* = 0) z_0 + (1-h)T_F) d\Psi(z_0). \end{aligned} \quad (35)$$

Firms' R&D costs, $R_F(t)$, can be expressed by

$$R_F(t) = \frac{1-h}{1+\delta} \nu(t)Q(t) \int_{z_0} e_F^{\delta+1} dz_0. \quad (36)$$

The growth is from a single source—innovation. By the law of large numbers, aggregate growth can be written as:

$$\begin{aligned} g(t) = \frac{\dot{Q}(t)}{Q(t)} = & \int_{z_0 \in \{z_0 | \tilde{q}^* > 0\}} (h\lambda_0 e_I(z_0, \tilde{q}^*) x_0(z_0, \tilde{q}^*) + (1-h)\lambda_0 e_F \kappa x_{F0}) d\Psi(z_0) \\ & + \int_{z_0 \in \{z_0 | \tilde{q}^* = 0\}} (h\lambda_0 e_I(z_0, \tilde{q}^* = 0) z_0 + (1-h)\lambda_0 e_F \kappa x_{F0}) d\Psi(z_0). \end{aligned} \quad (37)$$

The aggregate growth rate only depends on the allocation of inventor-led innovations across firms, which is invariant to time or $\nu(t)$, implying that:

$$g(t) = g.$$

Note that the growth rate of the aggregate quality can be decomposed into two parts:

$$\frac{\dot{Q}(t)}{Q(t)} = \frac{\dot{N}_F(t)}{N_F(t)} + \frac{Q(t)/N_F(t)}{Q(t)/N_F(t)} \quad (38)$$

¹⁹See Appendix C.4 for a formal proof.

The first term captures growth from the extensive margin (firms' mass), and the second term captures growth from the intensive margin (average quality).

3.8 Equilibrium

This section characterizes an equilibrium of the economy in which the aggregate variables (Y, C, R, w, Q) grow at a rate g , while the measure of firms and the distribution of firm quality evolve from given initial conditions.

Definition An equilibrium of this economy is the mapping between $\tilde{q}^*(t)$ and z_0 , the allocation $\left(\left\{y_j^*(t)\right\}_j, Y^*(t), C_I^*(t), C_H^*(t)\right)$, the prices $\left(w^*(t), r^*(t), \left\{p_j^*(t)\right\}_j\right)$, the growth rate g^* , the entry rate $\lambda_I^*(t)$, the measure of firms $N_F^*(t)$, and the firm quality distribution $f_t^*(\tilde{q})$, such that (1) for any $j \in [0, 1]$, $y_j^*(t)$ and $p_j^*(t)$ satisfy Equation (29); (2) the wage $w^*(t)$ satisfies Equation (30); (3) the interest rate $r^*(t)$ satisfies Equation (28); (4) the measure of the intermediate producers $N_F^*(t)$ satisfies Equation (27); (5) the mapping between $\tilde{q}^*(t)$ and z_0 comes from inventors' optimal choices; (6) the entry rates $\lambda_I^*(t)$ satisfy Equation (25); (7) R&D spending $C_I^*(t)$ and $R_F^*(t)$ satisfy Equation (35) and Equation (36), respectively; (8) the aggregate output Y^* satisfies Equation (31); (9) the aggregate consumption C_H^* satisfies Equation (34); (10) the growth rate g satisfies Equation (37); and (11) given the initial conditions $N_F(0)$ and $f_0(\tilde{q})$, the distribution $f_t^*(\tilde{q})$ evolves according to the law of motion implied by exogenous exit at rate τ , entry at the quality draw $f_{I,t}(\tilde{q}_I)$, and the quality improvements generated by inventor-led and firm-led innovation.

The equilibrium is block recursive. The prices $\left(w^*(t), r^*(t), \left\{p_j^*(t)\right\}_j\right)$, the normalized firm value v^* , the contracts (a^*, T_I^*, T_F^*) , the effort levels (e_I, e_F) , the idea allocation $\tilde{q}^*(z_0)$, the entry rate λ_I^* , and the growth rate g^* are independent of the measure of firms $N_F(t)$ and of the firm quality distribution $f_t(\tilde{q})$. These objects exhibit no transitional dynamics. Taking them as given, $N_F(t)$ and $f_t(\tilde{q})$ have transitional dynamics and are determined by their laws of motion starting from initial conditions $(N_F(0), f_0(\tilde{q}))$.

Proposition 1. *Consider the above-described economy starting with initial conditions $Q(0) > 0$, $N_F(0) > 0$, and $f_0(\tilde{q})$. There exists a unique equilibrium. In this equilibrium, growth is always balanced, i.e., $g(t) = g^*, \forall t$. The aggregate quality, $Q(t)$, aggregate output, $Y(t)$, and aggregate consumptions, $C_H(t)$ and $C_I(t)$, all grow at the rate g^* .*

The proof of Proposition 1 is provided in Appendix C.5. In what follows, we omit the time subscript t whenever doing so does not create confusion.

3.9 Inventor's Choice in a Closed Form

To illustrate the intuition and main channels in the model, this section presents a simplified setup that characterizes the inventor's trade-off in closed form. One modification is introduced in this setup: the realized innovation quality, $xQ(t)$, is drawn from a distribution with mean $x_0(z_0, \tilde{q})Q(t)$ and second moment $e_I^{-1}x_0(z_0, \tilde{q})^2Q(t)^2$, instead of from a uniform distribution $U((1 - \phi)x_0(z_0, \tilde{q})\nu, (1 + \phi)x_0(z_0, \tilde{q})\nu)$. For startups, x and $x_0(z_0, \tilde{q})$ are replaced by z and z_0 , respectively. This modification captures an environment in which inventor effort reduces innovation uncertainty. The variance of the inventor's consumption becomes

$$\text{Var}(c_I(a, T_I, \tilde{q}, z_0)) = a^2 \nu^2 Q(t)^2 \left(\underbrace{\tau \tilde{q}^2}_{\text{Var}(\tilde{V}(\tilde{q}))} + \underbrace{\lambda_0 (x_0(z_0, \tilde{q}))^2}_{\text{Var}(\text{innovation})} \right).$$

Innovation-related uncertainty no longer depends on the effort level.

The parameter δ in the effort cost function, $R(e_I)$, is assumed to take the value of 1, consistent with empirical estimates in the literature (e.g., [Akçigit and Kerr \(2018\)](#)). Using backward induction, firms anticipate that the inventor will choose the following effort:

$$e_I = \lambda_0 a x_0(z_0, \tilde{q}).$$

A higher inventor equity share, a , or a higher expected innovation quality, $x_0(z_0, \tilde{q})$, induces greater inventor effort. In both cases, the marginal return to effort rises, making additional investment more attractive.²⁰

The firm's problem in Equation (17) yields,

$$a^* = \frac{1}{1 + 2 \frac{A}{\lambda_0} \left(\frac{\tau \tilde{q}^2}{\lambda_0 x_0(z_0, \tilde{q})^2} + 1 \right)} \quad (39)$$

When $b < 1$, the optimal equity share a^* decreases with firm size $\tilde{q}$. This reflects two opposing forces that jointly determine a^* : the commercialization value $x_0(z_0, \tilde{q})$ and the non-innovation shock, captured by the exit risk $\tau \tilde{q}^2$. Firm size affects these components in opposite directions. On the one hand, larger firms provide greater synergy, which raises $x_0(z_0, \tilde{q})$ and makes it more valuable to incentivize inventor effort, pushing a^* up. On the other hand, larger firms expose the inventor to more non-innovation risk, so offering less equity shields the inventor from uncertainty unrelated to innovation, pushing a^* down. The net effect depends on which channel dominates. When $b < 1$, as in the calibration,

²⁰The derivation of the optimal contract is provided in Appendix C.6.

the exit-risk channel dominates, and larger firms optimally offer a smaller equity share to the inventor.

Denote the optimal compensation scheme by (a^*, T_I^*) . Under Bertrand competition, the wage component T_I^* is pinned down by the firm's zero-profit condition,

$$T_I^* = (1 - a^*) \lambda_0 e_I x_0(z_0, \tilde{q}).$$

Upon reviewing all contracts, an inventor with idea quality z_0 chooses the firm $\tilde{q}$ she would like to work for. If joining an incumbent, the optimal firm size is

$$\tilde{q}^* = \left(\frac{(2A\lambda_0 + \lambda_0^2) (\gamma(z_0))^2 b}{2A\tau q_0^{2b} (1 - 2b)} \right)^{\frac{1}{2-2b}}.$$

Proposition 2. *When $b < 0.5$, there is positive assortative matching when ideas join incumbents: $\frac{\partial \tilde{q}^*}{\partial z_0} > 0$.²¹*

When b is low, synergy does not grow without bound as firm size increases, and the countervailing force of a lower equity share guarantees existence and uniqueness of the solution. The model predicts positive assortative matching among incumbent firms: higher-quality innovations are more likely to be developed in larger firms. This arises for two reasons. First, better ideas gain more from commercialization synergy. Second, they generate relatively more innovation-related uncertainty, which makes inventors less vulnerable to incentive problems.

Proposition 3. *When $b < \frac{\min(z_0^{-\eta})}{\min(z_0^{-\eta}) + \max(B^{-\eta})}$, there exists a cutoff $\bar{z}_0(B)$, such that all inventors with $z_0 < \bar{z}_0(B)$ join incumbent firms.²²*

Both incumbents' adoption frictions and synergies increase with idea quality. If synergy gains do not rise too sharply with firm size, i.e., $b < \frac{\min(z_0^{-\eta})}{\min(z_0^{-\eta}) + \max(B^{-\eta})}$, adoption frictions dominate, and inventors with high-quality ideas choose to start new businesses.

Proposition 4. *When $b < \frac{\min(z_0^{-\eta})}{\min(z_0^{-\eta}) + \max(B^{-\eta})}$, and z_0 is drawn from a distribution $\Psi_{new}(z_0)$ that first-order stochastically dominates $\Psi(z_0)$, so ideas are better on average, a larger share of inventors launch startups.²³*

²¹See Appendix C.7 for the proof.

²²See Appendix C.8 for the proof.

²³See Appendix C.9 for the proof.

Higher idea quality raises innovation output in both incumbents and startups. However, incumbents must integrate new ideas into an existing technology base, which lowers their realization potential, especially for high-quality ideas. As a result, when ideas are more frontier-shifting on average, as at the peak of a technology wave, starting up new firms becomes relatively more attractive.

Together, Propositions 2 and 4 deliver the paper’s central prediction: a technology wave lowers market concentration. A wave draws ideas from a stochastically better distribution, which reshapes the firm-size distribution through two forces. The first is entry. A better idea distribution induces more inventors to launch startups (Proposition 4), expanding the mass of firms. The second one is quality improvement, operating through both firm-led and inventor-led innovation. Firm-led innovation raises every firm’s quality by a common proportional rate, and leaves the HHI unchanged. The inventor-led component has an impact on market concentration since it features positive assortative matching. In total, a wave simultaneously raises entry and, by drawing inventors away from incumbents, weakens the positive assortative matching that channels the best ideas into the largest firms (Proposition 2); it therefore lowers concentration along both the extensive and the intensive margin.

4 Calibration

We calibrate the full model to target the average US economy from 1982 to 2016. Patents are used as a surrogate for innovations. An innovation’s idea quality, denoted by z_0 , and the realized value, $xvQ(t)$ ($zvQ(t)$ in the context of a startup), correspond to the patent’s citation (scientific importance) and the patent’s economic value, respectively. Additionally, we assume that the idea quality z_0 follows the Pareto distribution characterized by a scale factor z_m and a shape factor α .

4.1 Identification

Parameters in the model are categorized into two groups. The first group is calibrated using a prior information from the aggregate statistics or the literature. The second group is calibrated by estimation from the micro-level data or through the model. Table 3 reports the parameters in the first group, $(\rho, \beta, \tau, A, \delta)$. The discount rate, ρ , is set to 0.02 to match the average interest rate in the sample period. The production function quality share, β , is 0.109, following Akcigit and Kerr (2018). The firm exit rate, τ , is 0.09, targeting the average exit rate of firms during our sample period based on the Business Dynamics

Statistics (BDS).²⁴ The scale parameter of the innovation arrival rate, λ_0 , is normalized to 1.²⁵ The risk aversion parameter, A , and the effort cost elasticity, δ , are set to be 0.5 and 1, respectively, which are commonly used in the literature (Hall and Van Reenen, 2000).

Table 3: Parameter Values from a Priori Information

Parameter	Description	Value	Identification
ρ	Discount rate	0.02	Interest Rate
β	Production function quality share	0.109	Firm profitability
τ	Exo. exit rate	0.09	BDS
λ_0	Innovation arrival rate	1	Normalization
A	Risk aversion	0.5	Risk aversion
δ	Effort cost elasticity	1	Effort cost elasticity

Notes: This table shows parameter values from the literature or direct estimation.

We jointly calibrate the remaining parameters in the second group, $(\alpha, z_m, \phi, \eta, B, b, q_0, h, \kappa)$, using the minimum distance method inspired by Lentz and Mortensen (2008). The parameters, along with their corresponding moments are shown in Table 4.

Table 4: Parameters from the Minimum Distance Estimation

Para.	Description	Identification
α	Shape of idea quality distribution	SD/Mean of patent citations
z_m	Scale of idea quality distribution	Aggregate economic value of innovations
ϕ	Innovation value dispersion	SD/Mean of economic value cond. on cites
B	Maturity of technology	Tech Wave Index
η	Elasticity of substitution	Estimation of the economic-value function
b	Exponent of the synergy function	Estimation of the economic-value function
q_0	Scale of the synergy function	New-to-incumbent ratio
h	Share of inventor-led innovation	Firm size ratio by cite quartiles (Q4/Q1)
κ	Scale of the firm-led innovation	the aggregate growth rate

Notes: Parameters in this table are jointly calibrated to minimize the distance between the model and data moments.

Growth Rate—Innovation is the sole driver of growth in the model. The scale parameter governing firm-led innovation, κ , is central to firms' innovation intensity and thus important to aggregate growth. A higher κ increases the step size of firm-led innovation, raising the overall growth rate. We calibrate κ so that the model-implied aggregate growth rate equals 1.7%, matching the average annual U.S. growth rate in GDP

²⁴The BDS data is compiled from the Longitudinal Business Database (LBD) by the Census Bureau.

²⁵In the model, λ_0 enters only multiplicatively, either as $\lambda_0 x_0(z_0, \tilde{q})$ or as $\lambda_0 \mathbb{E} [x_0(z_0, \tilde{q})^2]$, which implies that λ_0 is not separately identified from $x_0(z_0, \tilde{q})$.

per capita from 1982 to 2016.²⁶

The S.D.-to-Mean Ratio of Patent Citations—This ratio captures the dispersion in patent citations observed in the data, which reflects the underlying dispersion in inventors' idea quality. The parameter α is the primary driver of this dispersion. Specifically, the standard deviation-to-mean ratio of the idea distribution is given by $\frac{1}{\sqrt{\alpha(\alpha-2)}}$. Although the patents recorded in the USPTO data represent only successful innovations—a selected subset of all ideas—the dispersion in patent citations remains heavily influenced by α . We construct the citation distribution by pooling all patents granted since 1976 and their corresponding citations recorded by the USPTO, and compute the standard deviation-to-mean ratio, which is approximately 2.784.

Aggregate Economic Value of Innovation—The economic value of innovations is modeled as a direct contribution to firm value. In the model, this value follows a uniform distribution with a mean determined by the underlying idea quality (scientific value), z_0 . Given α , the average idea quality is governed by the scale parameter of the Pareto distribution, z_m . Accordingly, z_m is calibrated to match the ratio of average economic value of patents to average firm value.²⁷ Following the methodology of Kogan et al. (2017), patent economic value is inferred from stock market reactions using an extended dataset (1926–2022) linking U.S. patents to CRSP and Compustat data. While the raw aggregate patent value is 0.0069 times the average firm value for public firms, applying the statistical model by Yang (2026) to account for the broader firm population adjusts this ratio to 0.027. To balance these estimates, we adopt a calibrated value of 0.008.

S.D.-to-Mean Ratio of Innovations' Economic Value Conditional on Citations—Conditional on idea quality, the economic value of innovation is uniformly distributed, implying a standard-deviation-to-mean ratio of $\frac{\phi}{\sqrt{3}}$. Using the same sample as in the calibration of z_m , we estimate the standard-deviation-to-mean ratio of patent economic value while controlling for patent citations. In the data, this ratio is approximately 0.416.

Estimation of the Function of Patents' Economic Values—Both the elasticity of substitution between the technology base and new ideas, η , and the regression coefficient of innovations' economic value on incumbent firm size, b , are estimated directly from the

²⁶We compute annual growth rates from the Federal Reserve Economic Data (FRED) as the trend component obtained by applying the HP filter to GDP per capita.

²⁷Mathematically, the model expresses the ratio as a function of the underlying idea-quality distribution,

$$\begin{aligned} \frac{\mathbb{E}(\text{patent value})}{\mathbb{E}(\text{firm value})} &= \frac{\int_{z_0 \in \{z_0 | \tilde{q}^* > 0\}} h \lambda_0 e_I(z_0, \tilde{q}^*) x_0(z_0, \tilde{q}^*) \nu Q}{\nu Q / N_F} + \frac{\int_{z_0 \in \{z_0 | \tilde{q}^* = 0\}} (1-h) \lambda_0 e_F \kappa x_{F0} \nu N_F \int q dF(q) d\Psi(z_0)}{\nu Q / N_F} \\ &= N_F \int_{z_0 \in \{z_0 | \tilde{q}^* > 0\}} h \lambda_0 e_I(z_0, \tilde{q}^*) x_0(z_0, \tilde{q}^*) + N_F \int_{z_0 \in \{z_0 | \tilde{q}^* = 0\}} (1-h) \lambda_0 e_F \kappa x_{F0} d\Psi(z_0). \end{aligned} \quad (40)$$

data using nonlinear least squares. For each patent i granted to firm j in year t , we observe its economic value $x_{ijt}\nu Q(t)$, the firm's quality q_{jt} , the average technology base (proxied by average backward citations per patent) B_t , and the patent's forward citations z_{ijt} . The structural model for the underlying patent value is

$$x_{ijt}\nu Q(t) = \alpha \left(\frac{q_{jt}}{Q(t)} \right)^b (B_t^\eta + z_{ijt}^\eta)^{\frac{1}{\eta}} \nu Q(t) \varepsilon_{ijt},$$

where b and η are parameters of interest and ε_{ijt} is assumed to be a multiplicative disturbance with $E[\ln \varepsilon_{ijt} \mid q_{jt}, B_t, z_{ijt}] = 0$.

Taking logarithms and absorbing constants into an intercept and year fixed effects, we obtain the estimating equation

$$\ln(x_{ijt}\nu Q(t)) = c + b \ln q_{jt} + \frac{1}{\eta} \ln (B_t^\eta + z_{ijt}^\eta) + \delta_t + u_{ijt},$$

where c collects α and ν , δ_t is a year fixed effect capturing $Q(t)$, and u_{ijt} is an error term satisfying $E[u_{ijt} \mid q_{ijt}, B_t, z_{ijt}, t] = 0$. Let the full parameter vector collect all unknowns,

$$\theta = (c, b, \eta, \{\delta_t\}_{t \in \mathcal{T}}),$$

and define the residual

$$u_{ijt}(\theta) = \ln(x_{ijt}\nu Q(t)) - c - b \ln q_{jt} - \frac{1}{\eta} \ln (B_t^\eta + z_{ijt}^\eta) - \delta_t.$$

We estimate θ by nonlinear least squares, i.e.,

$$\hat{\theta} = \arg \min_{\theta} S(\theta), \quad S(\theta) = \sum_{ijt} u_{ijt}(\theta)^2.$$

At the optimum $\hat{\theta}$, the first-order conditions with respect to every component of θ are satisfied:

$$\frac{\partial S(\theta)}{\partial \theta_k} = 2 \sum_i u_i(\theta) \frac{\partial u_i(\theta)}{\partial \theta_k} = 0, \quad k = 1, \dots, \dim(\theta).^{28}$$

²⁸These conditions can be interpreted as a system of moment restrictions. Defining the moment vector

$$g(W_{ijt}, \theta) = u_{ijt}(\theta) \frac{\partial u_{ijt}(\theta)}{\partial \theta},$$

where $\frac{\partial u_{ijt}(\theta)}{\partial \theta}$ is the gradient of the residual with respect to all components of θ , the population moment conditions are

$$E[g(W_{ijt}, \theta_0)] = 0.$$

Thus, the nonlinear least squares estimator $\hat{\theta}$ can be viewed as a just-identified GMM estimator that jointly

The elasticity η and the scale factor b , included in θ , are identified in the estimation.

Technological Wave Index—Technology waves are defined as the ratio of forward citations to the sum of forward and backward citations for all patents granted in a given year. In the model, the existing technology base, B , which enters the realization potential of new ideas, $\gamma(z_0) = (z_0^\eta + B^\eta)^{1/\eta}$, corresponds to the stock of backward citations, representing the maturity of the existing technologies. The value of B is calibrated so that the model-generated average forward-to-backward citation ratio, $\int z_0 d\Psi(z_0) / (B + \int z_0 d\Psi(z_0))$, matches the average Tech Wave Index between 1982 and 2016, where $\int z_0 d\Psi(z_0)$ corresponds to the total forward citations of available ideas.

New-to-Incumbent Ratio—The scale parameter in the synergy function, $\tilde{q}_0$, governs the payoff from contributing an idea to an incumbent relative to starting a new venture, and thus shapes inventors' choice between incumbents and startups. We calibrate $\tilde{q}_0$ to match the “New-to-Incumbent Ratio” defined in Section 2.2.1, targeting its average value over 1982–2016.

Firm Size Ratio: Fourth-to-First Quartile of Patent Citations—The model predicts that, conditional on joining an incumbent, the inventors with higher-quality ideas should match with larger firms. This positive sorting is weakened by firm-led innovation, since ideas are sold to incumbents of random size. As the share of firm-led innovation increases, the relationship between idea quality and firm size becomes flatter. To calibrate the share of inventor-led innovation, h , we compute in the model the average incumbent firm size by patent-citation quartile (restricting to patents developed within incumbents) and match the implied fourth-to-first quartile firm-size ratio to its empirical counterpart.²⁹

4.2 Calibration Results

Table 5 reports the model-generated moments and their counterparts in the data. Overall, the model matches the targeted moments closely. The resulting parameter values are reported in Table 6.

Our estimates suggest that, relative to startups, incumbents realize lower innovation

estimates $(c, b, \eta, \{\delta_t\}_{t \in \mathcal{T}})$ by enforcing the orthogonality conditions implied by the model.

²⁹Within incumbents, innovations originating from an idea of quality z_0 (with $z_0 \in \mathcal{Z}$) fall into two categories: (i) inventor-led innovation and (ii) firm-led innovation. The average size of incumbent firms that own patents with z_0 in a set $\mathcal{Z}$ is

$$\mathbb{E}_{z_0 \in \mathcal{Z}}(q) = \int_{z_0 \in \mathcal{Z}} h q^*(z_0) \lambda_I^*(z_0) d\Psi(z_0) + \int_{z_0 \in \mathcal{Z}} (1 - h) \lambda_F^*(z_0) \int q f(q, t) dq d\Psi(z_0).$$

Here, $\int q f(q, t) dq$ is the average incumbent firm size. It enters the firm-led term because firm-led ideas are allocated randomly across incumbents according to the cross-sectional firm-size distribution $f(q, t)$.

Table 5: Moments

Identification Moment	Data	Model
Growth rate	0.017	0.017
S.d.-to-mean ratio of patent citations	2.78	2.78
Aggregate innovation value	0.008	0.008
S.d.-to-mean ratio of innovation value cond. on citations	0.416	0.416
Elasticity of Substitution between past Knowledge Stock and New Ideas	-0.31	-0.31
Technology Wave Index	0.54	0.54
Regression coefficient of innovation value on firm size	0.32	0.32
New-to-incumbent ratio	0.054	0.054
Firm size ratio by fourth-to-first-quartile of citations	1.18	1.19

Notes: This table compares the moments generated from the calibrated model and the data. In general, the model-generated moments match the data well.

Table 6: Estimated Parameter Values

Parameter	Description	Value
α	Shape of idea quality distribution	2.10
z_m	Scale of idea quality distribution	0.086
ϕ	Dispersion of patent economic value	0.72
η	CES elasticity of substitution	-0.31
B	Technology base	0.13
b	Exponent of the synergy function	0.32
q_0	Denominator of the synergy function	9.2E-5
h	Share of inventor-led innovation	0.39
κ	Scale of the firm-led innovation	3.21

Notes: Parameters in this table are jointly calibrated to minimize the distance between the model and data moments.

value on average because of incomplete pass-through. In particular, the ratio, $\frac{\int \gamma(z_0) d\Psi(z_0)}{\int z_0 d\Psi(z_0)} = 0.090$, is substantially below one. At the same time, synergy increases value. The scale parameter in the synergy function is $q_0 = 9.2 \times 10^{-5}$, and the elasticity is 0.32, implying that an average-sized incumbent can generate roughly 18 times the commercialization value of a startup through synergy alone. Once incomplete pass-through is taken into account, this net advantage falls to about 1.61 times.

Technology waves shape idea allocation along both the extensive and intensive margins. The extensive margin determines how many inventors found new businesses, while the intensive margin governs the firm size an inventor selects when joining an incumbent. Figure 5 illustrates both margins by plotting the optimal firm-size choice across idea qualities in inventor-led innovation at the calibrated B level. A positive firm size corresponds to joining an incumbent, whereas zero indicates founding a new

business. The figure indicates positive assortative matching among inventors who join incumbents: higher-quality ideas are matched with larger firms. Once idea quality, z_0 , exceeds a cutoff, the inventor instead chooses to launch a startup. This pattern is consistent with the fact that patents produced by startups receive more citations on average than those produced by incumbents in our data.

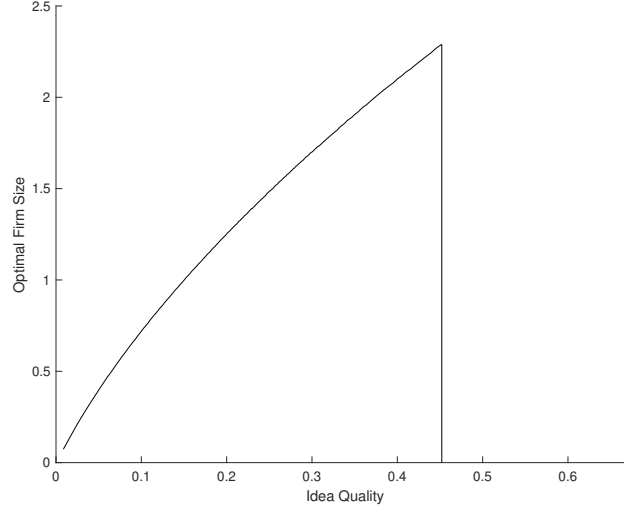


Figure 5: Firm-size Choices of Inventors

Notes: The figure shows the model-implied mapping from an inventor’s idea quality to the optimal firm size. A positive firm size corresponds to joining an incumbent of that size, whereas zero indicates founding a new business.

Variation in technology waves can be reflected as changes in the scale parameter of the idea-quality distribution, z_m , and the technology base parameter, B . A higher z_m raises average idea quality, encouraging startup formation and reducing the share of ideas subject to positive assortative matching within incumbents. A higher B increases the value of incumbents’ technology base, directing more ideas toward incumbents and expanding the role of assortative matching. Thus, z_m and B exert opposing effects on market concentration.

5 Quantitative Analysis

Using the calibrated model, we simulate the economy starting in 1984. Appendix E.3 reports analogous simulations starting in 1986, confirming that our results are robust to the choice of initial year. We choose z_{mt} and B_t to match the empirical Tech Wave Index each year, holding all other parameters fixed. Since the index identifies only their relative magnitudes, we hold $B_t + \frac{\alpha}{\alpha-1}z_{mt}$ —the sum of the technology stock and mean

idea quality—constant and normalize the sample mean of z_{mt} to its calibrated value z_m .³⁰

As an alternative specification, we vary only z_{mt} , holding B and all other parameters fixed.³¹ The simulations generate annual series for economic growth, market concentration, and the allocation of innovation. We compare these model-implied series with their empirical counterparts and report the corresponding correlations.

The initial firm-size distribution is pinned down by the empirical distribution from the Business Dynamics Statistics (BDS). Appendix E.1 describes the procedure in detail. Because the equilibrium features no transition dynamics, the evolution of firm mass $N_F(t)$ and the firm-size distribution $f_t(q)$ can be computed directly. The resulting paths of $N_F(t)$ and $f_t(q)$ jointly determine the time series of growth rates and market concentration.

5.1 Growth Rate

The aggregate growth rate, g , can be decomposed into growth in average firm quality and growth in firm mass, as shown in Equation (38). Using the baseline simulation, in which both z_m and B vary, we calculate five-year averages of aggregate growth and its two components. Figure 6 compares these series with five-year averages of the HP-filtered annual growth rate in the data. Two patterns stand out. First, firm mass contracts during 2000–2010, indicating negative net firm entry. Second, average firm quality behaves differently: growth in average firm quality is relatively strong in 2000–2010, consistent with faster improvements among surviving firms, but it is offset by a pronounced contraction in firm mass. These qualitative patterns align with the evidence in Autor et al. (2020) and Akcigit and Ates (2023): since the early 2000s, productive incumbents have expanded while net firm entry and overall dynamism have declined. The corresponding decomposition from the simulation in which only z_m varies is reported in Appendix E.4 and exhibits similar patterns.

5.2 Market Concentration

Figure 7 presents the simulated evolution of market concentration, measured by the Herfindahl–Hirschman Index (HHI), together with its empirical counterparts. Appendix

³⁰These restrictions imply

$$z_{mt} = \frac{z_m}{\overline{\text{TWI}}} \text{TWI}_t, \quad B_t = \frac{\alpha}{\alpha - 1} \frac{z_m}{\overline{\text{TWI}}} (1 - \text{TWI}_t),$$

where $\overline{\text{TWI}}$ is the sample mean. The constant-sum normalization isolates changes in the relative importance of new ideas and the existing technology base.

³¹We choose z_{mt} so that the model-implied Tech Wave Index, $\frac{\int z_0 d\Psi_t(z_0)}{B + \int z_0 d\Psi_t(z_0)}$, matches its empirical counterpart each year.

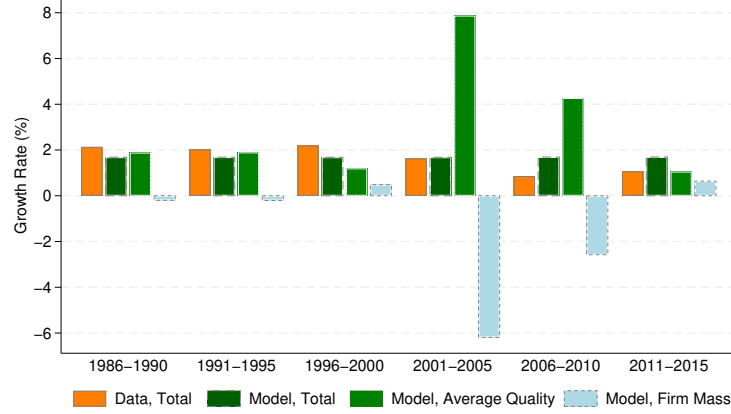


Figure 6: Decomposition of Growth Rates by Period

Notes: This figure reports 5-year averages of the aggregate economic growth rate and its two components. In each 5-year window, the left bar shows total growth. The middle bar shows the contribution from improvements in average firm quality, and the right bar shows the contribution from changes in firm mass.

E.2 details the construction of the model-simulated HHI. The solid red line shows the baseline simulated HHI with varying z_m and B , the dash-dotted red line shows the simulated HHI when varying only z_m , the dashed blue line reports the employment-based HHI from Census data, and the long dash-dotted blue line reports the sales-based HHI from Compustat. All series are normalized by their mean over 1984–2016. The model-generated HHI closely tracks both concentration measures in the data.

Although the calibration does not explicitly target any measure of market concentration, the model successfully replicates both the upward trend and cyclical fluctuations observed in the data. To disentangle these components, we fit separate linear trends to the simulated and empirical HHI series and subtract them to obtain the detrended time variations, as shown in Figure 8. Summary statistics are reported in the first three rows of Panel A in Table 7.

The fitted linear trend of the baseline HHI has a slope of 0.013, close to the empirical slopes of 0.009 and 0.012. The standard deviation of its detrended component is 0.091, nearly identical to the employment-based estimate of 0.090. Its first-order autocorrelation is 0.904, compared with 0.846 in the employment-based data and 0.928 in the sales-based data. The detrended baseline HHI has correlations of 0.552 and 0.835 with the employment- and sales-based series, respectively.

When only z_m varies, the model continues to reproduce these patterns. The fitted trend has a slope of 0.008, while the standard deviation of the detrended series is 0.069. Its autocorrelation is 0.908, and its correlations with the employment- and sales-based HHI measures are 0.560 and 0.832, respectively. Thus, variation in z_m alone accounts

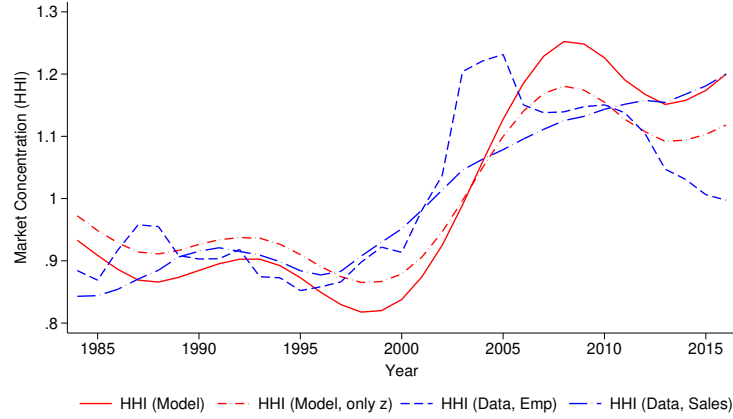


Figure 7: Model-generated HHI and Empirical Counterparts

Notes: The figure compares model-generated and empirical market concentration. The solid and dash-dotted red lines show the simulated HHI in the baseline and z_m -only cases, respectively, while the dashed and long dash-dotted blue lines show the employment-based LBD and sales-based Compustat measures. Each series is normalized by its 1984–2016 mean.

for much of the persistence and empirical comovement generated by the baseline model, while allowing B to vary improves the model's fit to the trend and volatility of market concentration. Taken together, these results indicate that technology waves are important for both the long-run trend and cyclical fluctuations in market concentration.

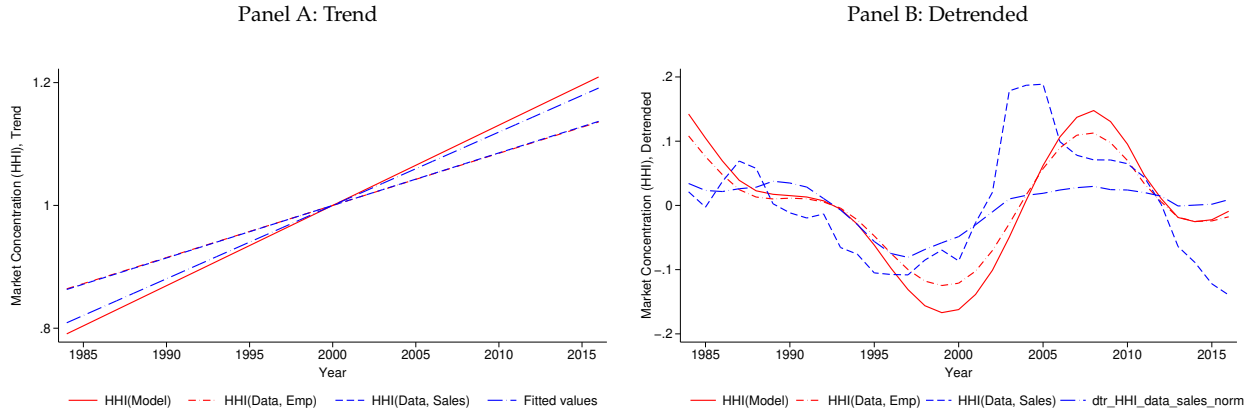


Figure 8: The Trend and Detrended Time Variations of the HHI

Notes: The left panel plots the linear trend in market concentration (HHI) over time in the model (red lines) and in the data (blue lines). The right panel shows the corresponding detrended HHI series, highlighting fluctuations around the trend.

Table 7: Comparison between Model and Data

	Original		Detrended		
	Mean	Slope	S.D.	Autocorr	Corr. Data
Panel A. HHI					
Data (Emp, Census)	1	0.009	0.090	0.846	1 (0.611)
Data (Sales, Compustat)	1	0.012	0.035	0.928	0.611 (1)
Model (z_m and B Change)	1	0.013	0.091	0.904	0.552 (0.835)
Model (Only z_m Change)	1	0.008	0.069	0.908	0.560 (0.832)
Model (Counterfactual)	1.11	0.017	0.086	0.914	0.549 (0.835)
Panel B. New-to-Incumbent Ratio					
Data	0.054	-6.72E-4	0.007	0.847	1
Model (z_m and B Change)	0.064	-1.84E-3	0.023	0.890	0.711
Model (Only z_m Change)	0.056	-8.26E-4	0.012	0.907	0.670

Notes: This table summarizes the trend and detrended time-series variation in the HHI and the new-to-incumbent ratio in both the data and the model. The columns report the series mean, the slope of the fitted linear trend, the standard deviation of the detrended series, the first-order autocorrelation of the detrended series, and the correlation of the detrended series with its data counterpart. For the HHI, two correlations are reported: with the employment-based measure and, in parentheses, with the sales-based measure.

5.3 Decomposition of the Extensive and Intensive Margins

Market concentration is shaped by both the extensive margin (the mass of firms) and the intensive margin (matching between idea quality and firm size). We use the allocation of ideas and a counterfactual exercise to examine the contribution of each margin.

5.3.1 Extensive Margin: Allocation of Ideas

Idea allocation shapes the evolution of firm mass: inventors are more likely to form startups during technological breakthroughs and to join incumbents as technologies mature. Panel A of Figure 9 shows that both the baseline simulation (solid green) and the z_m -only simulation (dash-dotted green) track the empirical New-to-Incumbent ratio (dashed black).

Panel B of Table 7 reports the corresponding statistics for the New-to-Incumbent ratio. In the baseline simulation, its mean is 0.064, compared with 0.054 in the data. Its trend is more negative (-1.84×10^{-3} versus -6.72×10^{-4}), and its detrended series is more volatile (0.023 versus 0.007), with a slightly higher autocorrelation (0.890 versus 0.847). When only z_m varies, the mean (0.056), trend (-8.26×10^{-4}), and standard deviation (0.012) move closer to their empirical counterparts, although autocorrelation rises to 0.907. The correlations with the empirical detrended series are 0.711 in the baseline and 0.670 in the z_m -only simulation, suggesting that technology waves account

for a substantial share of the variation in firm mass. The model's excess volatility may reflect the absence of adjustment costs: short-lived inventors choose between startups and incumbents without regard to previous affiliations, causing idea allocation to respond immediately to aggregate shocks.

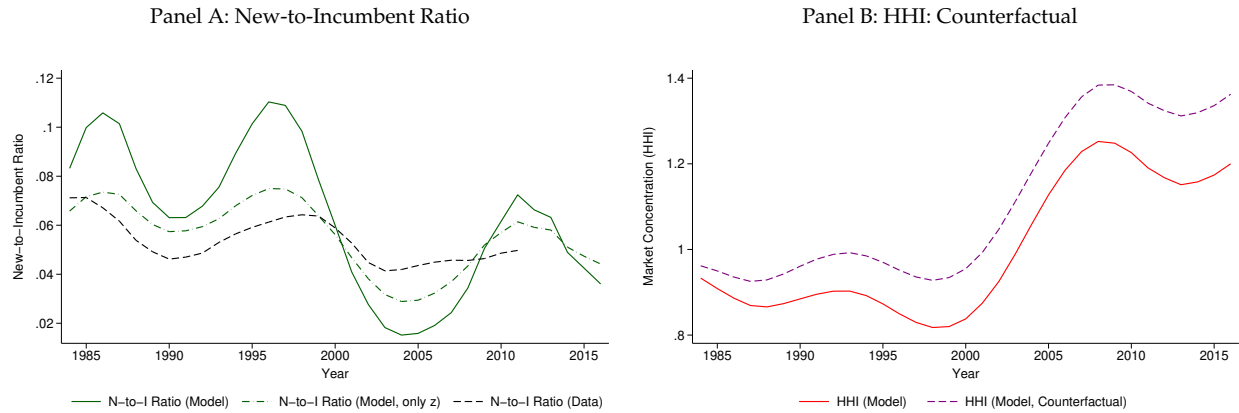


Figure 9: Idea Allocation and the Effect of PAM

Notes: Panel A plots the baseline and z_m -only simulated New-to-Incumbent ratios (solid and dash-dotted green lines) alongside their empirical counterpart (dashed black line; see Figure 2). Panel B isolates the intensive-margin contribution to model-implied market concentration by comparing the baseline HHI with a counterfactual in which new-firm entry is also governed by positive assortative matching. The counterfactual HHI is normalized by the mean of the baseline series.

5.3.2 Intensive Margin: Positive Assortative Matching

To isolate the intensive margin, we construct a counterfactual that preserves the baseline extensive margin period by period while extending assortative matching to startup ideas. The construction proceeds in five steps: (i) compute each inventor's optimal incumbent size; (ii) retain the baseline occupational choice and startup cutoff; (iii) leave the contracts, effort, and matching of inventors joining incumbents unchanged; (iv) preserve startup contracting, entry, and the entrant-size distribution; and (v) allow each startup to enter at its baseline drawn size without receiving the quality improvement generated by its inventor, and instead add that improvement to the incumbent of the inventor's optimal size. Thus, the counterfactual preserves the idea distribution, occupational choices, entry rate, firm mass, and growth rate, differing from the baseline only in the assignment of startup innovations across firms. Appendix E.5 provides the full five-step construction.

Panel B of Figure 9 compares the counterfactual HHI with the baseline, with both series normalized by the mean baseline HHI. Panel A of Table 7 reports the corresponding moments. The counterfactual raises average HHI by 11% and increases its trend slope from 0.013 to 0.017. Therefore, startup formation lowers both the level and trend slope of

HHI by attenuating positive assortative matching.

5.4 Relationship among Tech Waves, HHI, and Idea Allocation

This section examines the extent to which the model reproduces the relationships among technology waves, market concentration, and idea allocation observed in the data. We first estimate these relationships empirically using instrumental-variable regressions and then compare the resulting coefficients with their model-implied counterparts obtained from simulated data.

5.4.1 Tech Waves and Market Concentration

We examine the relationship between technology waves and market concentration using

$$\text{HHI}_{st} = \beta_0 \text{TWI}_{st} + \beta_1 \text{Size}_{st} + \theta_s + \mu_t + \epsilon_{st}, \quad (41)$$

where HHI_{st} is the Herfindahl–Hirschman Index for 6-digit NAICS industry s in year t . Industry size is measured by total employment or payroll to account for the relationship between market size and concentration (Campbell and Hopenhayn, 2005). The Tech Wave Index is constructed at the 4-digit IPC level and mapped to 6-digit NAICS industries using the industry composition of patent owners within each IPC class.

Because TWI may be correlated with unobserved concentration shocks or affected by reverse causality, we instrument it using lagged breakthrough intensity in technologically connected upstream IPC classes. The weights are determined by citation flows in the U.S. innovation network, following Acemoglu, Akcigit and Kerr (2016), Liu et al. (2025), and the shift-share framework of Borusyak, Hull and Jaravel (2022). The instrument is then mapped to NAICS industries using the same concordance. Appendix E.6 describes its construction in detail and reports the corresponding first-stage estimates and F-statistics in Table 12.

Table 8 reports the empirical 2SLS estimates and their model counterparts. Columns (1)–(2), (3)–(4), and (5)–(6) use contemporaneous, one-year-lagged, and two-year-lagged TWI, respectively; the second specification in each pair includes industry and year fixed effects. All six empirical estimates are negative.

Columns (7)–(9) report regressions using annual data generated by the quantitative model. We regress simulated HHI on contemporaneous, one-year-lagged, and two-year-lagged simulated TWI, using the same scaling as in the empirical analysis. The resulting coefficients are -0.119^{***} , -0.151^{***} and -0.172^{***} , compared with empirical 2SLS estimates of -0.241 to -0.306 , -0.238 to -0.367 , and -0.330 to -0.407 , respectively.

Table 8: Relationship between HHI and the Tech Wave Index: Empirical 2SLS Estimates and Quantitative Simulations

	Empirical estimates (2SLS)						Simulations (Model)		
	Contemporaneous		One-year lag		Two-year lag		Contemporaneous	One-year lag	Two-year lag
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
TWI	−0.241*** (0.0850)	−0.306** (0.139)	−0.238*** (0.0831)	−0.367** (0.148)	−0.330** (0.144)	−0.407*** (0.147)	−0.119*** (0.0356)	−0.151*** (0.0321)	−0.172*** (0.0291)
Observations	37,500	37,500	37,500	37,500	36,500	36,500	33	32	31
Industry size	Yes	Yes	Yes	Yes	Yes	Yes	N/A	N/A	N/A
Year FE	No	Yes	No	Yes	No	Yes	No	No	No
Industry FE	No	Yes	No	Yes	No	Yes	N/A	N/A	N/A

Notes: Columns (1)–(6) report empirical 2SLS estimates at the 6-digit NAICS-by-year level. Columns (1)–(2), (3)–(4), and (5)–(6) use contemporaneous, one-year-lagged, and two-year-lagged TWI, respectively. All empirical specifications control for industry size; standard errors are clustered at the 6-digit NAICS and year levels. The empirical observation counts are rounded to the nearest hundred to comply with Census Bureau disclosure requirements. Columns (7)–(9) report year-level regressions using data generated by the quantitative model, with normalized model HHI as the dependent variable and contemporaneous, one-year-lagged, or two-year-lagged simulated TWI as the explanatory variable. These specifications include a constant. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Thus, the model reproduces the negative relationship at each horizon, although its implied response is approximately one-third to one-half of the empirical magnitude.

5.4.2 Tech Waves and the New-to-Incumbent Ratio

The relationship between technology waves and startup decisions is estimated using

$$\text{NewFirm}_{it} = \beta_0 \text{TWI}_{c(i)t} + \beta_1 \ln(1 + \text{Citations}_{it}) + \theta_{c(i)} + \mu_t + \epsilon_{it}. \quad (42)$$

The dependent variable equals one if patent i is granted to a firm no more than five years old. TWI is measured at the 4-digit IPC-by-year level, and patent scientific value is proxied by forward citations received within five years. To address potential endogeneity, we instrument TWI using lagged technology waves in connected upstream IPC classes.³² The first-stage results are reported in Appendix E.6 and Table 12.

Columns (1)–(2) of Table 9 report the empirical 2SLS estimates without and with IPC and year fixed effects, respectively. The estimated TWI coefficients are 0.127 and 0.179, indicating a positive relationship between technology waves and the probability that an innovation is developed in a new firm.

Column (3) reports the corresponding quantitative-model regression. We simulate the startup decision for each idea and regress the resulting startup indicator on simulated

³²Because TWI is measured directly at the 4-digit IPC level, it does not need to be mapped to 6-digit NAICS industries.

Table 9: Starting New Businesses and the Tech Wave Index: Empirical 2SLS Estimates and Quantitative Simulations

	Empirical estimates (2SLS)		Simulation (Model)
	(1)	(2)	(3)
TWI	0.127*** (0.0477)	0.179*** (0.0514)	0.281*** (0.0022)
$\ln(1 + \text{Citations})$	0.00966*** (0.000935)	0.0108*** (0.000952)	
Initial idea quality (z_0)			0.557*** (0.0004)
Observations	4,032,000	4,032,000	4,032,000
IPC FE	No	Yes	N/A
Year FE	No	Yes	No

Notes: Columns (1)–(2) report empirical patent-level 2SLS estimates. Column (1) includes no fixed effects, while column (2) includes both 4-digit IPC and year fixed effects. Standard errors in these columns are clustered at the 4-digit IPC and year levels. Column (3) reports a linear probability regression using data generated by the quantitative model; the dependent variable is the simulated startup indicator, and the regression controls for initial idea quality z_0 . The simulated sample contains the same number of observations as the empirical sample. All specifications include a constant. Standard errors are reported in parentheses. The empirical observation counts are rounded to the nearest thousand to comply with Census Bureau disclosure requirements. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TWI, controlling for initial idea quality z_0 . The simulated sample is constructed to match the size of the empirical sample. The model-implied coefficient is 0.281, reproducing the positive empirical relationship but generating approximately twice its estimated magnitude.

6 Conclusion

This paper studies how technology waves shape growth and market concentration through the reallocation of ideas. Combining empirical evidence with a structural analysis, we show that market concentration is inversely related to technology waves, pointing to a low-frequency business-cycle component in the economy. Using the Longitudinal Business Database (LBD) from the U.S. Census Bureau and patent data from the USPTO, we document that the share of patents developed in new businesses comoves closely with technology waves: at wave peaks, a larger share of patents originates in new businesses, whereas at troughs, patenting shifts toward incumbents.

This paper proposes a theoretical framework that elucidates the decision-making process of inventors regarding their choice of innovation pathways, thus providing an

explanation for the observed empirical patterns. Our model effectively captures the contraction in firm mass and the increase in average firm quality after the 2000s, as well as both the trend and fluctuations in market concentration. It implies that the deceleration in the emergence of groundbreaking technologies could be a significant contributing factor to the decline in economic growth and the rise in market concentration after the 2000s.

The rise of generative AI may reduce market concentration by encouraging startup formation, consistent with the observed surge in AI business applications ([Dinlersoz, Dogan and Zolas \(2024\)](#)). The magnitude of this effect, however, depends on startups' reliance on incumbent platforms and products. How this technology wave shapes idea allocation and concentration remains a promising direction for future research.

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Online Appendix

A Data Description

The data used in this paper includes the Longitudinal Business Database (LBD), the USPTO patent data, the Compustat Fundamentals Annual, and PATSTAT Global (Spring 2024). This section provides details on the datasets and the construction of key variables.

A.1 The USPTO Patent Data

The USPTO patent data contain information on all patents issued between 1976 and 2022. It can be downloaded from the PatentsView website. For each patent, the data documents the patent type (utility, design, plant, etc.), the IPC code indicating its technological class, the grant year, and the patents it cites and it is cited. We keep all the utility patents to focus attention on the introduction of new products and processes.

Forward Citations Forward citations are citations a focal patent receives from others. It indicates how many patents follow the focal one. This paper calculates the number of forward citations each patent gets within five years after issuance.

Backward Citations Backward citations are citations that other patents receive from the focal patent. It indicates to what extent the focal patent follows the existing technology. This paper calculates the number of backward citations by counting the number of patents cited by the focal patent that were granted within the previous five years.

The Tech Wave Index According to the definition in the paper, we calculate this index by dividing the number of forward citations received by all the utility patents granted in a year by the summation of the forward and backward citations of those patents. The Tech Wave Index by IPC is derived in a similar way for each IPC class and each year.

A.2 The Compustat Fundamentals Annual

The Compustat Fundamentals Annual contains information of all the publicly listed firms in the US. It records the firms' net sales, the number of employees, the primary industry (4-digit SIC code), and the headquarter locations of each firm. We keep all the firms that are headquartered in the US.

Primary Industry The primary industry of each firm in Compustat is based on the 4-digit SIC code assigned to each firm in the Fundamentals Annual. The code can be aggregated to different levels. Manufacturing corresponds to SIC codes 2000-3999; utility and transportation corresponds to SIC codes 4000-4999; wholesale trade corresponds to

SIC codes 5000-5199; retail trade corresponds to SIC codes 5200-5999; finance corresponds to SIC codes 6000-6999; service corresponds to SIC codes 7000-8999.

The Herfindahl-Hirschman Index (HHI) Following the methods in [Grullon, Larkin and Michaely \(2019\)](#), we first calculate the HHI of each 3-digit SIC code by the squared ratios of firm net sales to the total net sales in that 3-digit industry. To get the aggregate HHI, we sum up the HHIs of all the 3-digit SIC codes and weight them by their total net sales.

A.3 The Longitudinal Business Database (LBD)

The LBD is collected by the US Census Bureau and is an establishment-level data that covers the universe of US businesses with paid employees from 1976 to 2020. The dataset assigns a firm ID to all establishments belonging to the same firm. Using the Business Dynamics Statistics of Patenting Firms (BDS-PF) patent assignee-FIRMID crosswalk from the Census, this paper links the USPTO patent data with firms in the LBD, therefore, derives all utility patents in the US that were granted to employer businesses between 1976 and 2020.

New-to-Incumbent Ratio After merging the patent data with the LBD, this paper can identify the firm each patent was granted to. If the firm is less than or equal to five years old in the patent's grant year, we indicate that the idea behind the patent was absorbed by a new firm 5 years ago. Otherwise, we indicate that the idea was absorbed by an incumbent firm 5 year ago. Then we divide the number of ideas combined with new firms by the number of ideas combined with incumbent firms to get the New-to-Incumbent Ratio.

Firm Size The LBD documents the number of employees each firm hires in each year. When deriving the mapping between patent forward citations and incumbent firm size, we use the number of employees as a proxy for size.

A.4 PATSTAT (Patent Statistical Database)

PATSTAT Global (Spring 2024) is a comprehensive global dataset maintained by the European Patent Office (EPO). PATSTAT provides detailed bibliographic data on patents from various patent offices worldwide, with a particular focus on those filed through the EPO. We restrict our sample to patents with inventors based in European countries.

Forward citations, backward citations, and the Tech Wave Index for European patents are constructed in the same way as for U.S. patents using USPTO data.

B More Empirical Evidence

This appendix presents additional evidence on technology waves, their relationship with idea allocation, and market concentration, following the sequence of the main text.

B.1 Technology Waves by Technological Field

The Tech Wave Index across the nine technological fields is shown in Figure 10. The index is based on the same algorithm as in Equation (1) except that the forward and backward citations are aggregated across each of the 1-digit IPC code. The top three fields with the highest Tech Wave Index are Human Necessities, Physics, and Electricity at the first peak; Electricity, Physics, and Human Necessities at the second peak; Human Necessities, Chemistry and Metallurgy, and Mechanical Engineering etc. at the third peak.

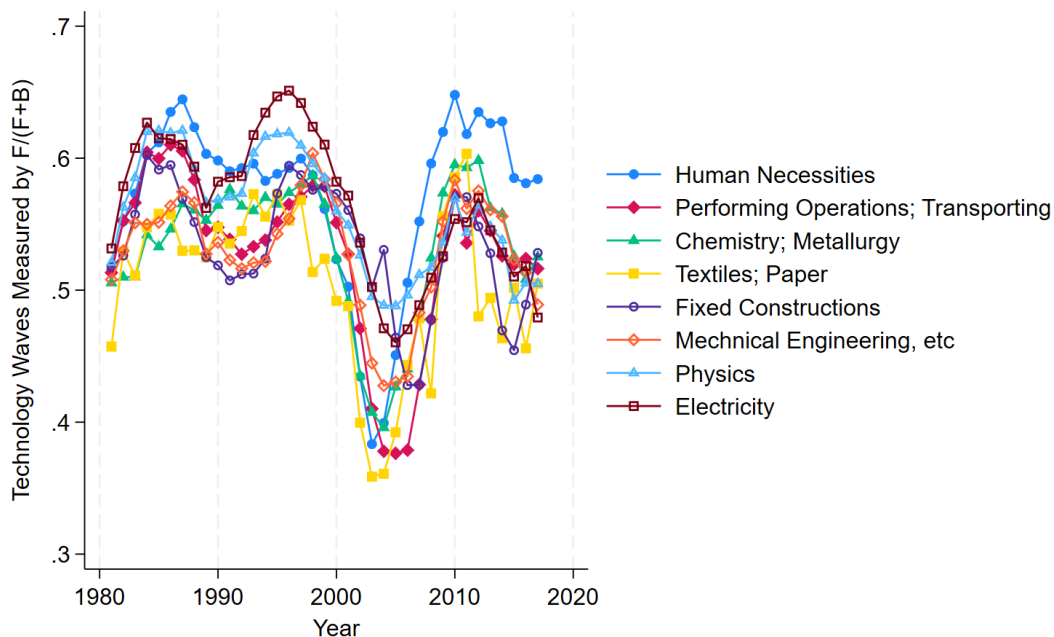


Figure 10: Technology Waves by Technological Fields

Notes: This figure shows the technology waves by the nine technological fields between 1981 and 2017. The nine fields are defined by the 1-digit IPC code. The technology waves are measured by the Tech Wave Index as defined by Equation 1 in the paper.

Sources: USPTO patent and citation data.

B.2 Main Drivers of the Wave Peaks

To capture the main technology classes that drive the peaks of the technology waves, we decompose the Tech Wave Index into the contribution of each three-digit IPC code using the following method,

$$TWI_t = \sum_{j \in J} \frac{\sum_{i \in I_{jt}} \sum_{s=0}^5 F_{ij,t+s}}{\sum_{i \in I_{jt}} \sum_{s=0}^5 F_{ij,t+s} + \sum_{i \in I_{jt}} \sum_{s=0}^5 B_{ij,t-s}} \frac{\sum_{i \in I_{jt}} \sum_{s=0}^5 F_{ij,t+s} + \sum_{i \in I_{jt}} \sum_{s=0}^5 B_{ij,t-s}}{\sum_{i \in I_t} \sum_{s=0}^5 F_{i,t+s} + \sum_{i \in I_t} \sum_{s=0}^5 B_{i,t-s}}, \quad (43)$$

where J is the set of 3-digit IPC code and I_{jt} is the set of patents belonging to the IPC code j granted in year t . Intuitively, the contribution of each technology class in a given year is determined by the IPC-specific Tech Wave Index multiplied by the share of forward and backward citations of that class. Table 10 lists the top three contributors at the three peaks of the technology waves. Medical or Veterinary Science and Hygiene contribute most to the first peak, while Computing; Calculating or Counting is the leading contributor to the second and third peaks.

Table 10: Major Contributors to the Peaks of Technology Waves

First Peak (1985-1987)	Second Peak (1995-1997)	Third Peak (2010-2012)
1 Medical or Vet. Sci.; Hygiene	Computing; Calculating or Counting	Computing; Calculating or Counting
2 Electric Elements	Medical or Vet. Sci.; Hygiene	Medical or Vet. Sci.; Hygiene
3 Measuring; Testing	Electric Communication Technique	Electric Communication Technique

Notes: This table shows the major technological classes of the top three fields with the highest Tech Wave Index at the peaks of technology waves in the period between 1981 and 2017.

B.3 Applicants of Breakthrough Patents

To identify the organizations behind technological breakthroughs, we draw on the “historically significant patents” compiled by Kelly et al. (2021) from online sources. Our sample period includes 54 such breakthrough patents. To assess whether these patents meaningfully contribute to aggregate technology waves, we compute a patent-level Tech Wave Index using the same methodology applied to the IPC-level index.³³ We then rank each of the 54 patents within the unconditional distribution of Tech Wave Index scores over the sample period. The mean and median percentile ranks fall in the top 24% and top 7% of all patents, respectively, indicating a strong concordance between Kelly et al.

³³Namely,

$$TWI_t = \sum_{i \in I_t} \frac{\sum_{s=0}^5 F_{i,t+s}}{\sum_{i \in I_t} \sum_{s=0}^5 F_{i,t+s}} \times \frac{\sum_{s=0}^5 F_{i,t+s}}{\sum_{i \in I_t} \sum_{s=0}^5 F_{i,t+s} + \sum_{i \in I_t} \sum_{s=0}^5 B_{i,t-s}}.$$

(2021)’s list and our novelty measure. We categorize the applicants of these breakthrough patents by institution type and find that 41% are incumbent firms, 26% are startups, 20% are universities, and the remainder come from other institutions, as shown in Figure 11. The wide range of sources demonstrates that frontier-shifting technologies emerge across diverse organizational settings rather than from a single dominant sector.

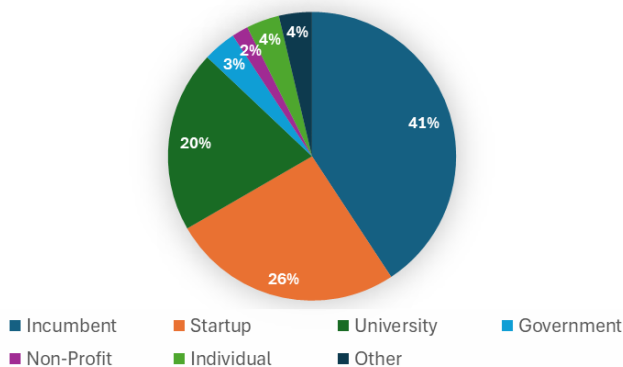


Figure 11: Composition of Applicants of Breakthrough Patents

Notes: The pie chart illustrates the share of applicants from each institutional group for the 54 breakthrough patents. Incumbents are defined as private firms that applied for a patent at least three years after their founding. Startups are private firms that applied within their first three years or before. Others include cases where the applicants are multiple institutions of different types.

Sources: The “historically significant patents” compiled by Kelly et al. (2021).

B.4 Tech Wave Index in Europe

To calculate the Tech Wave Index for European countries with intensive patenting activities, we use data from PATSTAT Global (Spring 2024). The six countries with the highest number of patent issuances between 1982 and 2016 are Germany, France, the United Kingdom, Italy, Switzerland, and the Netherlands. Using the definition of the Tech Wave Index as outlined in Equation (1), we calculate the technology waves in these six countries and present them in Figure 12. Across all six countries, we observe an overall declining trend in technological novelty, with common peaks in the mid-1980s and early 2010s. Italy also experienced a distinct peak in the mid-1990s. In general, the technological trends in these European countries with the highest patenting activity mirror those observed in the U.S.³⁴

³⁴Similar results can be obtained by calculating the Tech Wave Index using Google Patents data, as cleaned by Ayerst et al. (2023). These results are available upon request.

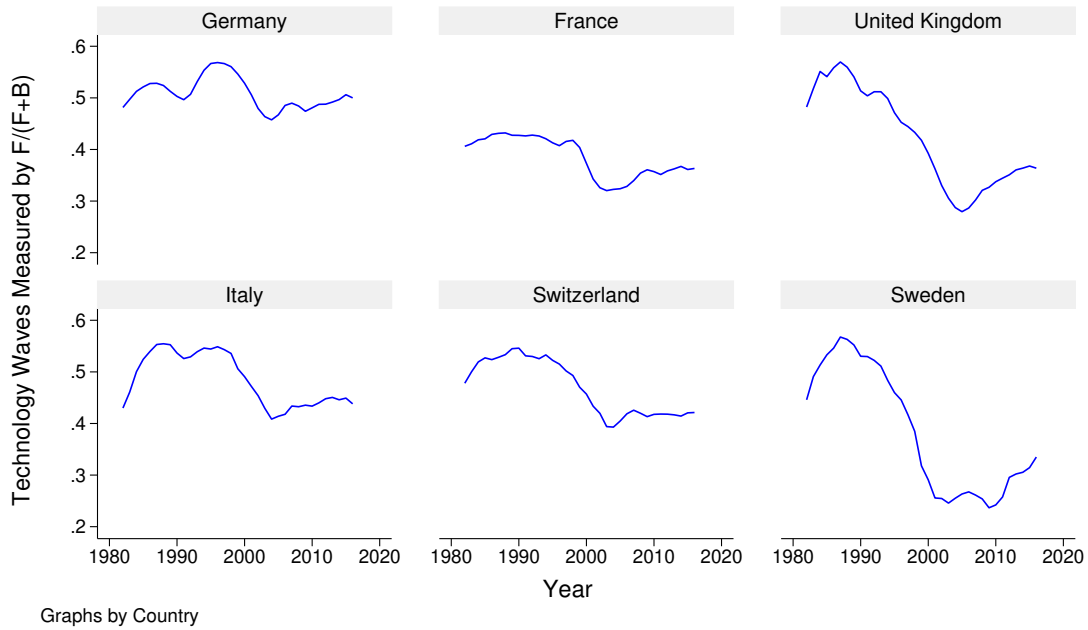


Figure 12: Technology Waves in European Countries

Notes: This figure shows the technology waves in six European countries with the highest number of patent issuances between 1982 and 2016. The technology waves are measured by the Tech Wave Index as defined by Equation 1 in the paper.

Sources: PATSTAT (Patent Statistical Database).

B.5 Idea Allocation without Smoothing

Figure 13 shows that the positive relationship between the Tech Wave Index and the New-to-Incumbent Ratio persists without smoothing. The corresponding evidence for market concentration appears in Appendix B.8.

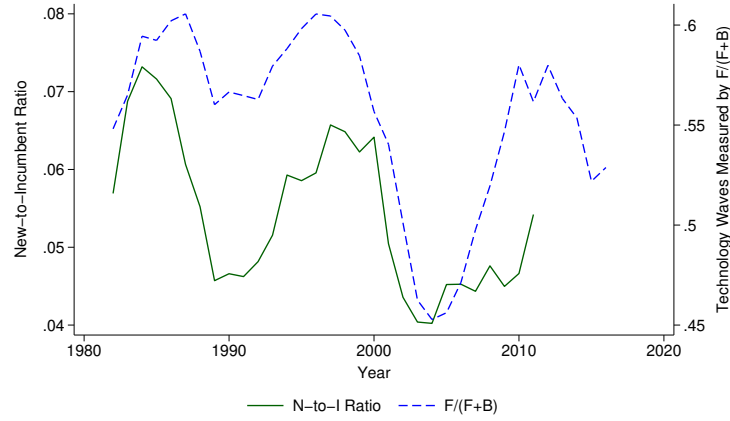


Figure 13: Technology Waves and Idea Allocation without Smoothing

Notes: This figure shows the technology waves and the idea allocation between new and incumbent firms over time. The blue dashed curve, based on the methodology defined in this paper, calculates the relative ratio of forward citations to the sum of forward and backward citations. The green solid curve displays the “New-to-Incumbent Ratio” defined in the paper, capture where new ideas contribute their value. The two curves have different y-axes, which are shown respective on the left and right.

Sources: Longitudinal Business Database (LBD) and USPTO patent and citation data.

B.6 IPC-Level Relationship between Tech Waves and New-to-Incumbent Ratio

We next examine the relationship between technology waves and idea allocation within technological fields, defined by the first digit of the IPC code. The IPC-level “Novelty” Index and “New-to-Incumbent Ratio” are computed using the same methodology as described in equations 1 and 3, with patent sets segregated according to their respective technology classes. Figure 14 illustrates that a positive correlation between idea allocation and technology waves is consistently observed across most technology classes. When a specific technology class experiences breakthroughs, there is an increase in the flow of ideas toward new startups. The contemporaneous correlation coefficients between the two curves are, respectively, 0.40 for Human Necessities, 0.30 for Performing Operations, 0.16 for Chemistry, 0.42 for Textiles, 0.39 for Fixed Constructions, -0.36 for Mechanical Engineering, 0.34 for Physics, and 0.58 for Electricity. The cross correlations when $k \in \{-3, -2, -1, 0, 1, 2, 3\}$ for each technological field are shown in Table 11 in Appendix B.7.

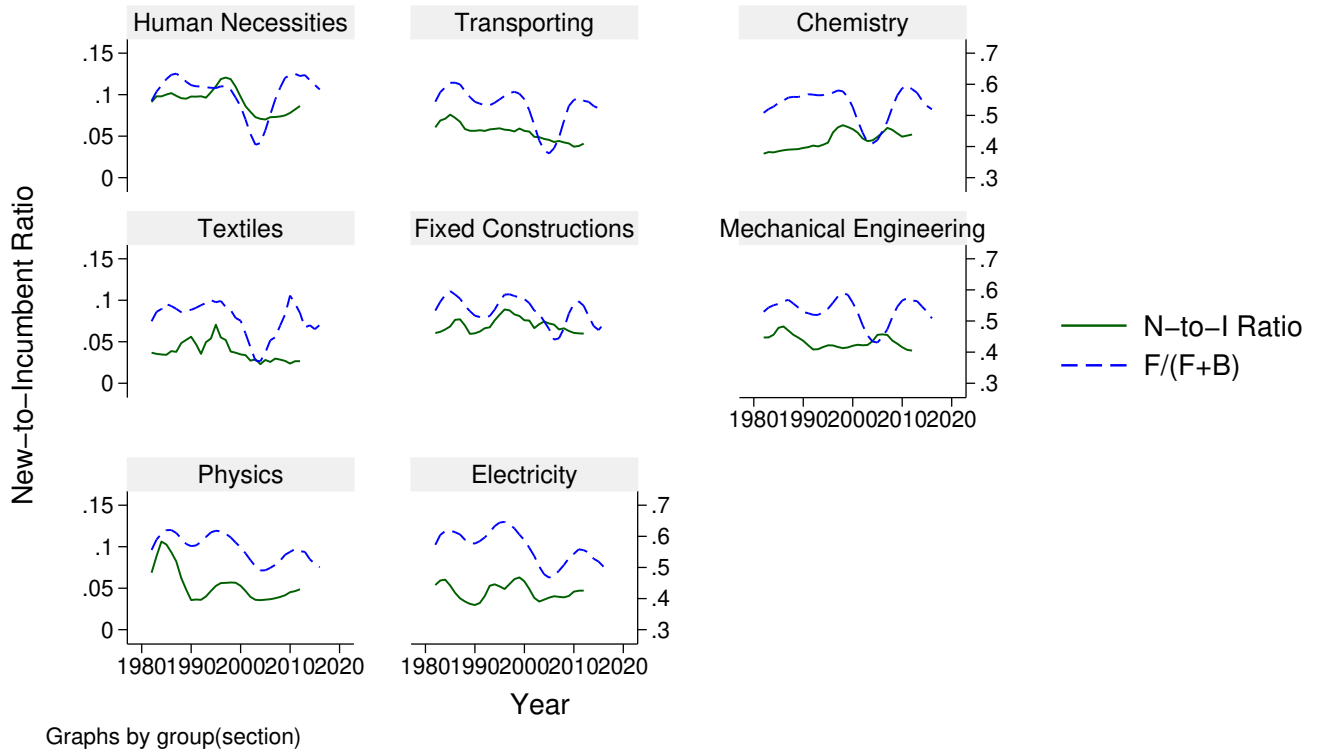


Figure 14: technology waves and Idea Allocation by Patent Technology Class

Notes: This figure shows the technology waves and the idea allocation between new and incumbent firms by patent technology class. The blue dashed curve, based on the methodology defined in this paper, calculates the relative ratio of forward citations to the sum of forward and backward citations. The green solid curve displays the “New-to-Incumbent Ratio” defined in the paper, capture where new ideas contribute their value. The two curves have different y-axes, shown respectively on the left and right.

Sources: Longitudinal Business Database (LBD) and USPTO patent and citation data.

B.7 Trends and Cross Correlations with Technology Waves

Table 11 summarizes trends and detrended cross correlations with technology waves for the New-to-Incumbent Ratio (Panel A) and the HHI (Panel B). Trends are slopes from linear regressions on year. Cross correlations are calculated after removing these trends, at year gaps $k \in \{-3, -2, -1, 0, 1, 2, 3\}$. Panel A reports aggregate and IPC-field results; Panel B reports aggregate concentration results.

Table 11: Time Trend and Cross Correlation

	Time Trend		Detrended Cross Correlation						
	Tech Wave	N-to-I Ratio	Panel A. New-to-Incumbent Ratio						
			$k = -3$	$k = -2$	$k = -1$	$k = 0$	$k = 1$	$k = 2$	$k = 3$
All	-0.002	-0.001	0.107	0.314	0.504	0.612	0.536	0.317	-0.001
Human Necessities	-0.001	-0.001	0.539	0.557	0.514	0.402	0.199	-0.063	-0.361
Performing Operations	-0.003	-0.001	0.117	0.210	0.283	0.301	0.191	0.017	-0.201
Chemistry; Metallurgy	-0.001	0.001	0.239	0.231	0.211	0.163	0.049	-0.128	-0.349
Textiles; Paper	-0.002	-0.001	0.458	0.451	0.462	0.417	0.371	0.270	0.189
Fixed Construction	-0.002	0	0.211	0.331	0.397	0.386	0.297	0.215	0.161
Mechanical Engineering	-0.001	0	-0.456	-0.505	-0.467	-0.358	-0.208	-0.061	0.059
Physics	-0.003	-0.001	-0.156	0.016	0.212	0.343	0.284	0.068	-0.207
Electricity	-0.003	-0.002	0.373	0.465	0.540	0.580	0.552	0.367	0.070
Panel B. HHI									
	Tech Wave	HHI	$k = -3$	$k = -2$	$k = -1$	$k = 0$	$k = 1$	$k = 2$	$k = 3$
All	-0.002	0.007	-0.625	-0.720	-0.775	-0.736	-0.588	-0.366	-0.097

Notes: This table shows the trends of the technology waves, HHI, New-to-Incumbent ratio and the detrended cross correlations among them. The trend is derived by running linear regressions of the focal time series on year and taking the coefficient; the cross correlations are derived by computing the correlation coefficients at different year gaps of the detrended time series.

B.8 Market Concentration without Smoothing

Figure 15 shows that the negative relationship between the Tech Wave Index and the HHI also persists without smoothing.

B.9 Alternative Measures of Market Concentration

To complement the HHI measures in the main text, we examine concentration using the receipt shares of the largest firms. This paper adopts the cleaned data series by [Kwon, Ma and Zimmermann \(2023\)](#) to compute the three-year moving average of the receipt share of the top 0.1% and 1% firms. The top shares are generated by the IRS data, which covers a more comprehensive set of firms. So, they can be used as a complement to the HHI measure in the paper. As displayed in Figure 16, the top shares exhibit increasing trends in general but with fluctuations. The peaks and troughs of the fluctuations appear nearly simultaneously with the HHI measured in this paper, showing the robustness of the market concentration patterns shown in the paper.

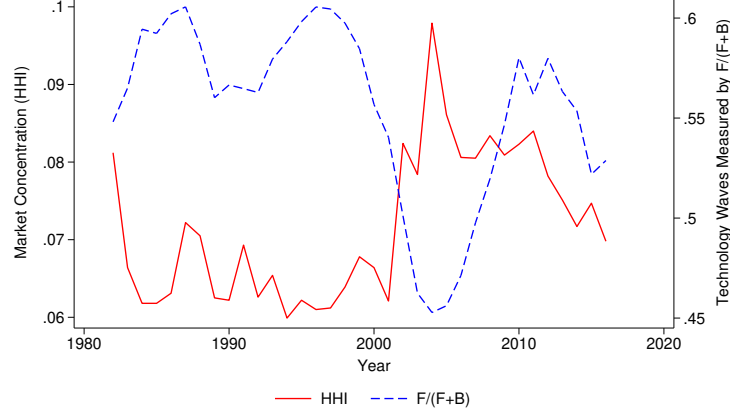


Figure 15: Technology Waves and Market Concentration without Smoothing

Notes: This figure shows the technology waves and market concentration of employment over time. The blue dashed curve, based on the methodology defined in this paper, calculates the relative ratio of forward citations to the sum of forward and backward citations. The red solid curve displays the HHI in each year, which is the weighted average of the industry-level HHI in each year. The weight is the total employment of firms in each industry. The two curves have different y-axes, shown respective on the left and right.

Sources: Compustat Fundamental Annuals and USPTO patent and citation data.

C Model and Proof

C.1 Starting up New Businesses

The partner's problem has the same form as the incumbent firm's, with $\tilde{q} = 0$ and the average innovation value being z_0 instead of $x_0(z_0, \tilde{q})$ (Equation (17)).

$$\begin{aligned}
 & \max_a (1-a) \lambda_0 e_I \int V(j, t | zQ(t)) dz - T_I V(t | Q(t)) \\
 & \text{st } e_I = \arg \max \{u(c_I(a, 0, T_I), e_I)\} \\
 & \quad u(c_I(a, T_I, 0, Q(t)), e_I) \geq \bar{u}(z_0) \\
 & \quad (1-a) \lambda_0 e_I \int V(j, t | zQ(t)) dz - T_I V(t | Q(t)) \geq 0
 \end{aligned} \tag{44}$$

The partners are assumed to get zero net payoff due to competition.

The inventor decides her effort level by maximizing her utility, which yields:

$$e_I = \lambda_0 a z_0 - A a^2 \lambda_0 \mathbb{E} \left(z(z_0)^2 \right) \tag{45}$$

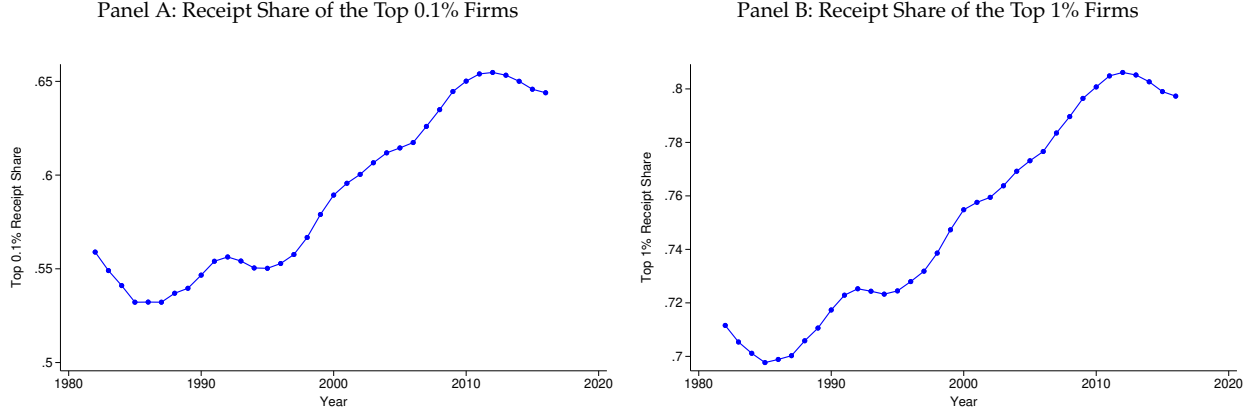


Figure 16: Receipt Shares of Top firms

Notes: This figure shows the three-year moving average of the receipt share of the top 0.1% (Panel A) and 1% firms (Panel B). The receipt shares are from the cleaned data series by [Kwon, Ma and Zimmermann \(2023\)](https://businessconcentration.com/), which is posted on <https://businessconcentration.com/>. The data source is the Statistics of Income (SOI) and the associated Corporation Source Book published annually by the IRS. Their statistics cover the whole population of US corporations.

Sources: <https://businessconcentration.com/>.

The firm's problem in Equation 44 becomes

$$\begin{aligned} \max_a & \lambda_0 e_I z_0 - A a^2 \left(\lambda_0 e_I \mathbb{E} \left(z(z_0)^2 \right) \right) - \frac{1}{2} e_I^2 \\ \text{st } & e_I = \lambda_0 a z_0 - A a^2 \lambda_0 \mathbb{E} \left(\tilde{z}(z_0)^2 \right) \end{aligned} \quad (46)$$

It gives the highest utility an inventor can obtain when working in a startup.

C.2 Production: Profit Maximization

The production sector features two types of firms: a representative final goods producer and intermediate goods producers. The final good producer assembles intermediate goods, denoted by j within the range $[0, N_F(t)]$, to produce final goods. The final goods producer chooses $\{y(t)_j\}_j$ to maximize its profit using the technology described in Section 3.2. The subscript t is omitted in this section whenever it does not cause a confusion. The final goods producer's problem can be written as:

$$\max_{\{y_j\}} \frac{1}{1-\beta} \int_0^{N_F} q_j^\beta y_j^{1-\beta} dj - \int_0^{N_F} y_j p_j dj. \quad (47)$$

The first-order condition

$$p_j = q_j^\beta y_j^{-\beta}$$

yields the demand function for goods produced by intermediate firms.

The intermediate goods are produced by their corresponding firm $j \in [0, N_F]$ using only labor $y_j = Ql_j$. Intermediate good producers engage in monopolistic competition, optimizing their profit by choosing l_j, p_j, y_j , given the wage level w :

$$\begin{aligned} \max_{l_j, p_j, y_j} & y_j p_j - w l_j. \\ \text{s.t. } & y_j = Q l_j \\ & p_j = q_j^\beta y_j^{-\beta} \end{aligned} \tag{48}$$

The FOC yields:

$$y_j(t) = q_j(t) \left(\frac{Q(t)(1-\beta)}{w(t)} \right)^{\frac{1}{\beta}}, l_j(t) = \frac{y_j(t)}{Q(t)}, p_j(t) = \frac{w(t)}{Q(t)(1-\beta)}.$$

The labor market clears, which derives that

$$\frac{\int_0^{N_F} q_j \left(\frac{Q(1-\beta)}{w} \right)^{\frac{1}{\beta}} dj}{Q} = 1$$

Therefore, the wage is $w = (1-\beta)Q$.

C.3 Firm-led Innovation

For firm-led innovation, plug the linear value function (Equation 33) into the firm choice problem specified in Equation 22

$$\begin{aligned} \Omega_F(z_0, q, t) &= \max_{e_F, T_F} \lambda_0 e_F \int \kappa x_F(z_0) v(t) N_F(t) q dx - \frac{1}{1+\delta} e_F^{\delta+1} v(t) N_F(t) q - T_F v(t) N_F(t) q \\ &= \max_{e_F, T_F} \left(\lambda_0 e_F \kappa x_{F0} - \frac{1}{1+\delta} e_F^{\delta+1} - T_F \right) v(t) N_F(t) q \end{aligned} \tag{49}$$

FOC yields:

$$e_F = (\lambda_0 \kappa x_{F0})^{\frac{1}{\delta}},$$

which is invariant to time and firm size. Equation 23 can be written as:

$$\lambda_0 e_F \kappa x_{F0} v(t) - \frac{1}{1+\delta} e_F^{\delta+1} v(t) = T_F v(t)$$

which yields:

$$\begin{aligned} T_F &= \lambda_0 e_F \kappa x_{F0} - \frac{1}{1+\delta} e_F^{\delta+1} \\ &= \frac{\delta}{1+\delta} (\lambda_0 \kappa x_{F0})^{\frac{1+\delta}{\delta}} \end{aligned} \quad (50)$$

which is also invariant to time and firm size. Inventor's utility in this case is:

$$U_I(T_F(z_0, q), 0) = T_F V(t|N_F(t)q) = \frac{\delta}{1+\delta} (\lambda_0 \kappa x_{F0})^{\frac{1+\delta}{\delta}} \nu(t) N_F(t) q,$$

which is linear in $\nu(t) N_F(t)$.

C.4 Inventor-led Innovation

In the inventor-led case, the consumption is

$$\begin{aligned} c_I(a, T_I, q, z_0, Q(t)) dt &= T_I V(t|Q(t)) dt + a dV(j, t|q, z_0), \\ &= T_I \nu(t) Q(t) dt \begin{cases} + a \tau \nu(t) q dt, & pr = 1 - \tau dt - \lambda_0 e_I dt \\ - a(1 - \tau dt - r dt) \nu(t) q, & pr = \tau dt \\ + a((1 - r dt) x(z_0, \tilde{q}) Q(t) + \tau q dt) \nu(t), & pr = \lambda_0 e_I dt \end{cases} \end{aligned} \quad (51)$$

which is linear in $\nu(t)$. The expected consumption is

$$\mathbb{E}(c_I(a, T_I, q, z_0, Q(t))) = (a \lambda_0 e_I x_0(z_0, \tilde{q}) + T_I) \nu(t) Q(t),$$

and the associated variance is

$$\begin{aligned} \text{Var}(c_I(a, T_I, \tilde{q}, z_0, Q(t)) dt) &= \mathbb{E}((c_I dt)^2) - \mathbb{E}(c_I dt)^2 \\ &= \mathbb{E}\left(a^2 \nu(t)^2 Q(t)^2 \left(\tilde{q}^2 \tau dt + x^2(z_0, \tilde{q}) \lambda_0 e_I dt\right)\right) \\ &\quad - \mathbb{E}(a \nu x_0(z_0, \tilde{q}) Q(t) \lambda_0 e_I dt)^2 \\ &= a^2 \nu(t)^2 Q(t)^2 \left(\tilde{q}^2 \tau + \mathbb{E}(x^2(z_0, \tilde{q})) \lambda_0 e_I\right) dt \end{aligned}$$

The variance comes from two sources: non-innovation-related firm value and the R&D process. Both terms increase in firm size $\tilde{q}$, but the former one increases in a faster speed, implying that in larger firms, shocks unrelated to R&D are stronger. Hence, larger firms are subject to larger incentive problems and the equity held by the inventor provides a

weaker incentive for R&D efforts. Upon reviewing all available contracts, an inventor determines her preferred firm $\tilde{q}$.

The instant utility is:

$$U_I(c_I(t), e_I(t)) = \left((a\lambda_0 e_I x_0(z_0, \tilde{q}) + T_I) - Aa^2 \left(\tilde{q}^2 \tau + \mathbb{E}(x^2(z_0, \tilde{q})) \lambda_0 e_I \right) - R(e_I) \right) v(t) Q(t) \quad (52)$$

The inventor's problem is to maximize her utility:

$$\max_{e_I(t)} U_I(c_I(a, T_I, q, Q(t)), e_I(t)).$$

FOC yields:

$$e_I(t) = \left(a\lambda_0 x_0(z_0, \tilde{q}) - Aa^2 \mathbb{E}(x^2(z_0, \tilde{q})) \lambda_0 \right)^{\frac{1}{\delta}} \quad (53)$$

which is invariant to $v(t)$.

The wage rate T_I is determined by the Bertrand competition:

$$(1-a) \lambda_0 e_I \left[\int V(j, t|q + x(z_0, \tilde{q}(t)) Q(t)) dx - V(j, t|q) \right] - T_I V(t|Q(t)) = 0$$

$$T_I = (1-a) \lambda_0 e_I x_0(z_0, \tilde{q}(t))$$

which is invariant to $v(t)$. Therefore, the instant utility is linear in $v(t)Q(t)$:

$$\begin{aligned} U_I(c_I(t), e_I(t)) &= \left((a\lambda_0 e_I x_0(z_0, \tilde{q}) + T_I) - Aa^2 \left(\tilde{q}^2 \tau + \mathbb{E}(x^2(z_0, \tilde{q})) \lambda_0 e_I \right) - R(e_I) \right) v(t) Q(t) \\ &= \left(\lambda_0 \left(\left(1 - \frac{a}{\delta+1} \right) x_0(z_0, \tilde{q}) - \frac{\delta}{\delta+1} Aa^2 \mathbb{E}(x^2(z_0, \tilde{q})) \right) e_I - Aa^2 \tilde{q}^2 \tau \right) v(t) Q(t) \\ &= u_I(z_0, \tilde{q}) v(t) Q(t) \end{aligned} \quad (54)$$

where $u_I(z_0, \tilde{q}) = \left(\lambda_0 \left(\left(1 - \frac{a}{\delta+1} \right) x_0(z_0, \tilde{q}) - \frac{\delta}{\delta+1} Aa^2 \mathbb{E}(x^2(z_0, \tilde{q})) \right) e_I - Aa^2 \tilde{q}^2 \tau \right)$, invariant to $v(t)$.

The inventor's choice of firm, as specified in Equation (24), can be written as:

$$u(z_0, t) = \max\{u_I(z_0, \tilde{q}^*) v(t) Q(t), u_I(z_0, 0) v(t) Q(t)\} \quad (55)$$

Since $v(t)Q(t) > 0$, the decision rule is the same as:

$$u(z_0, t) = \max\{u_I(z_0, \tilde{q}^*), u_I(z_0, 0)\}. \quad (56)$$

Thus, the decision rule $\tilde{q}^*(z_0)$ does not change over time.

C.5 Proof of Proposition 1

Proof. Given the linear value function $V(j, t \mid q) = v(t)q$ in (33), the inventor's effort e_I , the equity share a , the wage T_I , and the firm's effort $e_F = (\lambda_0 \kappa x_{F0})^{1/\delta}$ are all invariant to $v(t)$, and the inventor's utility in every option is proportional to the common factor $v(t)Q(t)$; consequently the decision rule $\tilde{q}^*(z_0)$ depends on parameters alone and is constant over time (Section 3 and Appendix C.4). Because the aggregate growth rate (37) is an integral over this allocation, $g(t) = g$ for all t (Section 3, Aggregate Variables).

The fraction h plays no role in this comparison. It is an exogenous, idea-level attribute: a share $1 - h$ of ideas is best implemented entirely by incumbents and is sold at the transfer T_F , while the remaining share h is inventor-led. The partition is realized before any choice is made, so h shifts the measure of inventors who face the startup–incumbent comparison but not the comparison itself. Even if each inventor were subject to the lottery ex ante, the firm-led payoff $T_F v(t) N_F(t) q = \frac{\delta}{1+\delta} (\lambda_0 \kappa \gamma(z_0))^{\frac{1+\delta}{\delta}} v(t) N_F(t) q$ does not depend on $\tilde{q}^*$: it enters her objective additively and drops out of the argmax. Hence h affects aggregate outcomes—the growth rate (37), the entry rate (25), and inventor consumption (35)—but never the individual decision rule.

Because $\tilde{q}^*(\cdot)$ is a fixed function of parameters, the partition of the support of Ψ into the incumbent region $\{z_0 : \tilde{q}^*(z_0) > 0\}$ and the startup region $\{z_0 : \tilde{q}^*(z_0) = 0\}$ is parameter-determined as well. Substituting $e_F = (\lambda_0 \kappa \gamma(z_0))^{1/\delta}$ into (37), the firm-led component collapses to a pure parameter integral, and

$$g^* = h \int \lambda_0 e_I(z_0, \tilde{q}^*(z_0)) x_0(z_0, \tilde{q}^*(z_0)) d\Psi(z_0) + (1 - h) \lambda_0^{\frac{1+\delta}{\delta}} \int (\kappa \gamma(z_0))^{\frac{1+\delta}{\delta}} d\Psi(z_0), \quad (57)$$

with $x_0(z_0, 0) = z_0$ in the startup region. In the general model e_I is

$$e_I(z_0, \tilde{q}) = \lambda_0 a x_0(z_0, \tilde{q}) - A a^2 \lambda_0 \mathbb{E} \left(x(z_0, \tilde{q})^2 \right),$$

and the optimal equity share a is characterized only numerically, so e_I has no explicit expression. It is nevertheless bounded. Because the variance term enters negatively and $a \in [0, 1]$,

$$0 \leq e_I(z_0, \tilde{q}) \leq \lambda_0 a x_0(z_0, \tilde{q}) \leq \lambda_0 x_0(z_0, \tilde{q}), \quad (58)$$

so effort in the general model never exceeds the closed-form effort level $\lambda_0 a x_0$ of Section C.6. Substituting (58) into (57), the inventor-led integrand obeys $h \lambda_0 e_I(z_0, \tilde{q}^*) x_0(z_0, \tilde{q}^*) \leq h \lambda_0^2 x_0(z_0, \tilde{q}^*)^2$, and it remains only to check that this is integrable on each region. On the incumbent region $\{z_0 \leq \bar{z}_0\}$, which is bounded above by the cutoff of Proposition 3, $x_0(z_0, \tilde{q}^*(z_0))$ is continuous in z_0 on a bounded set so the

integral is finite. On the startup region $\{z_0 > \bar{z}_0\}$ we have $x_0 = z_0$, and the envelope $h\lambda_0^2 z_0^2$ is integrable if and only if $\int z_0^2 d\Psi(z_0) < \infty$, which holds for the Pareto distribution used in the calibration provided $\alpha > 2$. Similarly, under the same condition, the firm-led term is also finite. Hence $g^* \in (0, \infty)$ is a constant determined by parameters and Ψ alone; it is well defined for every $h \in [0, 1]$.

Production is linear in aggregate quality, so $\dot{Y}/Y = \dot{Q}/Q = g^*$. Define

$$s_H(t) \equiv \frac{C_H(t)}{Y(t)}, \quad s_I v(t) \equiv \frac{C_I(t)}{Y(t)}, \quad s_F v(t) \equiv \frac{R_F(t)}{Y(t)}.$$

The coefficients $s_I, s_F \geq 0$ are constants, since all normalized policies are time-invariant, and $C_I(t)$ and $R_F(t)$ are linear in $v(t)Q(t)$ by (35) and (36). The resource constraint then gives $s_H(t) = 1 - (s_I + s_F)v(t)$.

Household optimization implies the Euler equation

$$\frac{\dot{C}_H(t)}{C_H(t)} = r(t) - \rho.$$

Because $C_H(t) = Y(t)s_H(t)$, we have

$$\frac{\dot{C}_H(t)}{C_H(t)} = \frac{\dot{Y}(t)}{Y(t)} + \frac{\dot{s}_H(t)}{s_H(t)} \tag{59}$$

$$r(t) = \rho + g - \frac{(s_I + s_F)\dot{v}(t)}{1 - (s_I + s_F)v(t)} \tag{60}$$

Plugging $V(j, t | q) = v(t)q$ into the HJB yields:

$$\dot{v}(t) = (r(t) + \tau)v(t) - \beta. \tag{61}$$

Substitute $r(t)$ from (60) to obtain

$$\dot{v}(t) = ((\rho + \tau + g)v(t) - \beta)(1 - (s_I + s_F)v(t)). \tag{62}$$

which is an autonomous scalar ODE, where $(s_I + s_F) \geq 0$ and $s_H(t) = 1 - (s_I + s_F)v(t) > 0$. v is always within the range $(0, \frac{1}{(s_I + s_F)})$. From Equation (62), it is straightforward that the only non-explosive point within the feasible range is $v(t) = v^*$.

With v constant, $s_H = 1 - (s_I + s_F)v^*$ is constant, $\dot{s}_H = 0$, and (60) gives $r(t) = \rho + g^*$. Because $C_H(t) = s_H Y(t)$, $C_I(t) = s_I v^* Y(t)$, $R_F(t) = s_F v^* Y(t)$ and $Y(t) \propto Q(t)$, all aggregates grow at the common rate g^* .

□

C.6 Closed-Form Model Solution

The firm's problem in Equation 17 can be rewritten as:

$$\begin{aligned} \max_a & \left(\lambda_0 e_I x_0(z_0, \tilde{q}) - A a^2 \left(\tau \tilde{q}^2 + \lambda_0 x_0(z_0, \tilde{q})^2 \right) - \frac{1}{2} e_I^2 \right) \nu Q \\ \text{st } e_I &= \lambda_0 a x_0(z_0, \tilde{q}) \end{aligned} \quad (63)$$

Putting the expression of e_I into the maximization problem and taking the FOC with regard to the equity share, a , derives,

$$\begin{aligned} a^* &= \frac{\lambda_0^2 x_0(z_0, \tilde{q})^2 \nu^2}{\lambda_0^2 x_0(z_0, \tilde{q})^2 \nu^2 + 2A \left(\tau \tilde{q}^2 \nu^2 + \lambda_0 x_0(z_0, \tilde{q})^2 \nu^2 \right)} \\ &= \frac{1}{1 + 2 \frac{A}{\lambda_0} \left(\frac{\tau \tilde{q}^2}{\lambda_0 x_0(z_0, \tilde{q})^2} + 1 \right)} \end{aligned} \quad (64)$$

Upon reviewing all contracts, an inventor with idea quality z_0 chooses which firm $\tilde{q}$ to work for by maximizing her instant utility:

$$\begin{aligned} \max_{\tilde{q}} u(c_I, e_I) &= \mathbb{E}(c_I) - \frac{A}{\nu Q(t)} \text{Var}(c_I) - R(e_I) \nu Q(t) \\ \text{st } a &= a^*(\tilde{q}) \\ T_I &= T_I^*(\tilde{q}) \end{aligned} \quad (65)$$

Putting the expression of the optimal equity level, $a^*(\tilde{q})$, and $T_I^*(\tilde{q}) = (1 - a^*) \lambda_0 e_I x_0(z_0, \tilde{q})$ into the maximization problem and solving the first-order condition,

$$\frac{\partial x_0(z_0, \tilde{q})}{\partial \tilde{q}} = \frac{2A\tau}{4A\tau \frac{\tilde{q}}{x_0(z_0, \tilde{q})} + \frac{2A\lambda_0 + \lambda_0^2}{\tilde{q}/x_0(z_0, \tilde{q})}}, \quad (66)$$

derive the optimal firm size,

$$\tilde{q}^* = \left(\frac{(2A\lambda_0 + \lambda_0^2) (\gamma(z_0))^2 b}{2A\tau q_0^{2b} (1 - 2b)} \right)^{\frac{1}{2-2b}}.$$

The left-hand side and right-hand side of the first-order condition when $b < 0.5$ and $\eta = -1$ are shown in Figure 17.

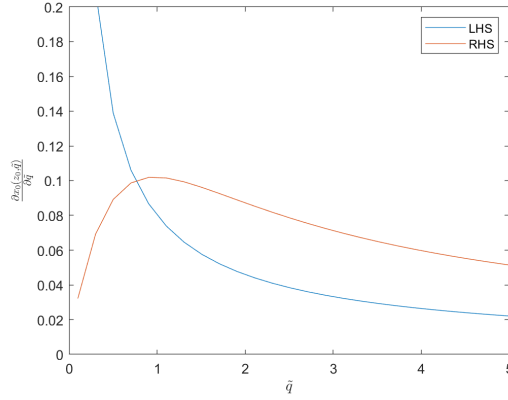


Figure 17: FOC Condition

Notes: This figure shows respectively the left-hand side (the blue curve) and the right-hand side (the red curve) of Equation 66. When $b < 0.5$, there exists a unique intersection.

C.7 Proof for Proposition 2

Proof.

$$\begin{aligned} \frac{\partial \tilde{q}^*}{\partial z_0} &= \frac{\tilde{q}^*}{\gamma(z_0)(1-b)} \frac{\partial (\gamma(z_0))}{\partial z_0} \\ &= \frac{\tilde{q}^*}{\gamma(z_0)(1-b)} (z_0^\eta + B^\eta)^{\frac{1}{\eta}-1} z_0^{\eta-1} \end{aligned}$$

where

$$\tilde{q}^* = \left(\frac{(2A\lambda_0 + \lambda_0^2)(\gamma(z_0))^2 b}{2A\tau q_0^{2b}(1-2b)} \right)^{\frac{1}{2-2b}}.$$

When $b < 0.5$, the optimal size increases in the inventor's idea quality, z_0 . □

C.8 Proof of Proposition 3

Proof. An inventor decides whether to launch a startup by comparing the highest utility offered by incumbents $u_I(z_0)$ and startups $u_N(z_0)$. When $\frac{\gamma(z_0)^{\frac{1}{1-b}}}{z_0} < \left(\frac{\lambda_0}{\tilde{q}_0(\lambda_0 + 2A)} \right)^{\frac{1}{2}}$, $u_I(z_0) < u_N(z_0)$, the inventor chooses to initiate a startup. When $\eta < 0$, if $b < \frac{\min(z_0^{-\eta})}{\min(z_0^{-\eta}) + \max(B^{-\eta})}$, $b - \frac{B^\eta}{z_0^\eta + B^\eta} < 0$ always holds:

$$\frac{\partial \left(\gamma(z_0)^{\frac{1}{1-b}} z_0^{-1} \right)}{\partial z_0} = \frac{\gamma(z_0)^{\frac{1}{1-b}}}{z_0^2} \frac{1}{1-b} \left(b - \frac{B^\eta}{z_0^\eta + B^\eta} \right) < 0,$$

since $\gamma(z_0) = (z_0^\eta + B^\eta)^{\frac{1}{\eta}}$. $\gamma(z_0)^{\frac{1}{1-b}} z_0^{-1}$ monotonically decreases in z_0 , when holding B constant. It implies there exists a cutoff $\bar{z}_0(B)$, when $z_0 > \bar{z}_0(B)$,

$$\frac{\gamma(z_0)^{\frac{1}{1-b}}}{z_0} < \left(\frac{\lambda_0}{\hat{q}_0(\lambda_0 + 2A)} \right)^{\frac{1}{2}}$$

always holds, and hence $u_I(z_0) < u_N(z_0)$, the inventors opts for new businesses instead of incumbent firms. $\square$

C.9 Proof for Proposition 4

Proof. By Proposition 3 the occupational choice is a threshold rule in own idea quality: inventor z_0 chooses a startup if and only if $z_0 > \bar{z}_0(B)$. The threshold does not depend on the idea distribution.

Let $\Psi_{\text{new}}(z_0)$ first-order stochastically dominate $\Psi(z_0)$. The share of inventors who launch startups is $h(1 - \Psi(\bar{z}_0(B)))$, and first-order stochastic dominance implies $\Psi_{\text{new}}(\bar{z}_0(B)) \leq \Psi(\bar{z}_0(B))$. Hence

$$h(1 - \Psi_{\text{new}}(\bar{z}_0(B))) \geq h(1 - \Psi(\bar{z}_0(B))), \quad (67)$$

with strict inequality whenever the dominance is strict at $\bar{z}_0(B)$: when ideas are better on average, a larger share of inventors start new businesses. $\square$

C.10 Full Model

The firm's problem in Equation 17 becomes

$$\begin{aligned} \max_a \left(\lambda_0 e_I x_0(z_0, \tilde{q}) - Aa^2 \left(\tau \tilde{q}^2 + \lambda_0 e_I \mathbb{E} \left(x(z_0, \tilde{q})^2 \right) \right) - \frac{1}{2} e_I^2 \right) \nu Q \\ \text{st } e_I = \lambda_0 a x_0(z_0, \tilde{q}) - Aa^2 \lambda_0 \mathbb{E} \left(x(z_0, \tilde{q})^2 \right) \end{aligned} \quad (68)$$

FOC yields the optimal a solves a cubic equation ³⁵:

$$FOC = 2A^2\lambda_0^2E(x^2)^2a^3 - \left(\lambda_0^2E(x)^2 + 2A\lambda_0^2E(x)E(x^2) + 2A\tau q^2\right)a + \lambda_0^2E(x)^2 = 0$$

Characterize the optimal allocation a^* . The solutions to the FOC is $a_1 \leq a_2 \leq a_3$. When $a \rightarrow -\infty, f(a) < 0$, and $a = 0, f(a) > 0$. Therefore, there is at least one solution satisfying $a_1 < 0$. Consider the SOC. It can be written in this form: $f'(a) = C_1a^2 + C_2, C_1 = 6A^2\lambda_0^2E(x(z_0, \tilde{q}))^2 > 0, C_2 = (\lambda_0^2E(x(z_0, \tilde{q}))^2 + 2A\lambda_0^2E(x(z_0, \tilde{q}))E(x(z_0, \tilde{q}))^2 + 2A\tau\tilde{q}^2) > 0$. Obviously, there exists a cutoff $\bar{a} > 0$, such that $SOC > 0$ when $a \in (-\infty, -\bar{a}) \cup (\bar{a}, \infty)$, and $SOC \leq 0$ when $a \in [-\bar{a}, \bar{a}]$. Therefore, there is one and only one solution $a_1 < 0$, and there can be either two positive solutions $a_2, a_3 > 0$ (including the case where $a_2 = a_3$), or two solutions with a imaginary imaginary. We are interested in the case where $a^* \in (0, 1)$ and $SOC < 0$, which is a_2 (the smaller one between the two positive solutions), if it exists (easy to show that SOC at a_3 is always positive). Consider four cases:

1. $a_2 \in [0, 1], a_3 > 1$. This is the optimal allocation. $a^* = a_2$.
2. $a_2 \in [0, 1], a_3 < 1$. a_2 is an interior solution and $a = 1$ is a boundary case that may or may not be dominated by a_2 . Therefore, we need to compare the case where $a = 1$ with a_2 to determine a^* .
3. $a_2 \notin R$, meaning that $FOC > 0, \forall a > 0$. $a^* = 1$.
4. $a_2 > 1, FOC > 0, \forall a \in [0, 1]$. $a^* = 1$.

Given the contracts, the inventor chooses which firm $\tilde{q}$ to work for by maximizing her utility, given her utility level.

$$\begin{aligned} & \max_{\tilde{q}} u(c_I(a, T_I, q, Q(t)), e_I) \\ & = \left(a\lambda_0 e_I x_0(z_0, \tilde{q}) + T_I - Aa^2 \left(\tilde{q}^2 \tau + \mathbb{E}(x^2(z_0, \tilde{q})) \lambda_0 e_I \right) - R(e_I) \right) \nu Q(t) \\ & \text{st } e_I = \lambda_0 a(z_0, \tilde{q}) x_0(z_0, \tilde{q}) - Aa(z_0, \tilde{q})^2 \lambda_0 \mathbb{E}(x(z_0, \tilde{q}) (z_0, \tilde{q})^2) \end{aligned}$$

³⁵In startup, the decision rule becomes:

$$FOC = 2A^2E(x^2)^2a^3 - \left(E(x)^2 + 2AE(x)E(x^2)\right)a + E(x)^2 = 0$$

$$FOC = 2A^2k_z^2z^2a^3 - (1 + 2Ak_zz)a + 1 = 0$$

C.11 Alternative

An alternative specification embeds firm size inside the CES aggregator:

$$x_0(z_0, \tilde{q}(t)) = q_0 \left((\tilde{q}(t)^b B)^\eta + z_0^\eta \right)^{1/\eta}, \quad \eta < 1, b > 0, \quad (69)$$

where q_0 is a parameter. Here firm size enters through the effective technology base $\tilde{q}(t)^b B$ rather than as a separate multiplicative synergy term. This functional form generates model features similar to those of the baseline model. All closed-form propositions still hold under the same or weaker assumptions. Proofs are below. The Proposition 1 does not rely on the functional form of x_0 . It hold under the alternative functional form assumption.

Differentiating $\ln x_0 = \ln q_0 + \frac{1}{\eta} \ln [(\tilde{q}^b B)^\eta + z_0^\eta]$:

$$\frac{\partial \ln x_0}{\partial \ln \tilde{q}} = \frac{1}{\eta} \cdot \frac{\eta b (\tilde{q}^b B)^\eta}{(\tilde{q}^b B)^\eta + z_0^\eta} = b \left(\frac{q_0 \tilde{q}^b B}{x_0} \right)^\eta \in (0, b), \quad (70)$$

$$\frac{\partial \ln x_0}{\partial \ln z_0} = \frac{1}{\eta} \cdot \frac{\eta z_0^\eta}{(\tilde{q}^b B)^\eta + z_0^\eta} = \left(\frac{q_0 z_0}{x_0} \right)^\eta \in (0, 1), \quad (71)$$

$$\frac{\partial \ln x_0}{\partial \ln B} = \left(\frac{q_0 \tilde{q}^b B}{x_0} \right)^\eta \in (0, 1). \quad (72)$$

Larger firms still commercialize better, and better ideas are still worth more.

Differentiating again,

$$\frac{\partial^2 \ln(x_0)}{\partial \ln \tilde{q}^2} = b \frac{\partial}{\partial \ln \tilde{q}} \left(\frac{q_0 \tilde{q}^b B}{x_0} \right)^\eta = \eta b^2 \left(\frac{q_0 \tilde{q}^b B}{x_0} \right)^\eta \left(\frac{q_0 z_0}{x_0} \right)^\eta, \quad (73)$$

$$\frac{\partial^2 \ln(x_0)}{\partial \ln z_0 \partial \ln \tilde{q}} = b \frac{\partial}{\partial \ln z_0} \left(\frac{q_0 \tilde{q}^b B}{x_0} \right)^\eta = -\eta b \left(\frac{q_0 \tilde{q}^b B}{x_0} \right)^\eta \left(\frac{q_0 z_0}{x_0} \right)^\eta. \quad (74)$$

When $\eta < 0$, the $\frac{\partial \ln x_0}{\partial \ln \tilde{q}}$ falls with firm size and rises with idea quality.

Under the alternative functional form assumption, Equation 63 to Equation 66 still hold, while the optimal firm size does not yield a closed form any more:

$$e_I = \lambda_0 a x_0, \quad a^* = \frac{1}{1 + \frac{2A}{\lambda_0} \left(\frac{\tau \tilde{q}^2}{\lambda_0 x_0^2} + 1 \right)}, \quad T_I^* = (1 - a^*) \lambda_0 e_I x_0. \quad (75)$$

As in the baseline case, a^* falls with $\tilde{q}$ if and only if $\tilde{q}^2/x_0^2$ rises with $\tilde{q}$, which holds when

$b < 1$.

The inventor's utility in a firm size $\tilde{q}$ is:

$$u(z_0, \tilde{q}) = \frac{1}{2} a^* \lambda_0^2 x_0^2 \nu Q = \frac{\lambda_0^4}{2} \cdot \frac{x_0^4}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \nu Q. \quad (76)$$

Taking logs in (76) and differentiating with respect to $\ln \tilde{q}$, using (70),

$$\frac{\partial \ln u(z_0, \tilde{q})}{\partial \ln \tilde{q}} = 2 \left[\frac{\partial \ln x_0}{\partial \ln \tilde{q}} \left(1 + \frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \right) - \frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \right]. \quad (77)$$

This is equivalent to Equation (66).

Lemma 1. *At any interior solution $\tilde{q}^*$ of (66), $\frac{\partial \ln x_0}{\partial \ln \tilde{q}} \big|_{\tilde{q}=\tilde{q}^*} < \frac{1}{2}$.*

Proof. When $\frac{\partial \ln u(z_0, \tilde{q})}{\partial \ln \tilde{q}} = 0$, the following holds:

$$\frac{\partial \ln x_0}{\partial \ln \tilde{q}} \left(1 + \frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \right) = \frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2}$$

It is easy to see that $\frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \in (0, 1)$. Therefore, $\frac{\partial \ln x_0}{\partial \ln \tilde{q}} \big|_{\tilde{q}=\tilde{q}^*} < \frac{1}{2}$. $\square$

Lemma 2. *Let $\eta < 0$ and $b > 0$. For every $z_0 > 0$, Equation (66) has a unique solution $\tilde{q}^*(z_0) \in (0, \infty)$, and it is the global maximizer of $u(z_0, \tilde{q})$.*

Proof. Differentiate (77) with respect to $\ln \tilde{q}$. Using (73) for the first factor and, for the risk term,

$$\frac{\partial}{\partial \ln \tilde{q}} \left[\frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \right] = \frac{4A\tau\tilde{q}^2 (\lambda_0^2 + 2A\lambda_0)x_0^2}{[(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2]^2} \left[1 - \frac{\partial \ln x_0}{\partial \ln \tilde{q}} \right], \quad (78)$$

one obtains

$$\begin{aligned} \frac{\partial^2 \ln u_I}{\partial (\ln \tilde{q})^2} &= 2 \frac{\partial^2 \ln(x_0)}{\partial \ln \tilde{q}^2} \left[1 + \frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \right] \\ &\quad - \frac{8A\tau\tilde{q}^2 (\lambda_0^2 + 2A\lambda_0)x_0^2}{[(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2]^2} \left[1 - \frac{\partial \ln x_0}{\partial \ln \tilde{q}} \right]^2. \end{aligned} \quad (79)$$

For $\eta < 0$ the first line is strictly negative and the second is non-positive, so $\partial \ln u_I / \partial \ln \tilde{q}$ in (77) is strictly decreasing in $\ln \tilde{q}$. Therefore, the $u_I(z_0, \cdot)$ is strictly log-concave in $\ln \tilde{q}$ and has at most one critical point, necessarily a global maximum.

As $\tilde{q} \rightarrow 0^+$, $(\tilde{q}^b B)^\eta \rightarrow +\infty$, so $\frac{\partial \ln x_0}{\partial \ln \tilde{q}} = b(q_0 \tilde{q}^b B / x_0)^\eta \rightarrow b$. $x_0 \rightarrow q_0 z_0 > 0$, and $\left[\frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \right] \rightarrow 0$. $\frac{\partial \ln u(z_0, \tilde{q})}{\partial \ln \tilde{q}} \big|_{\tilde{q}=0} > 0$.

As $\tilde{q} \rightarrow \infty$, $\frac{\partial \ln x_0}{\partial \ln \tilde{q}} \rightarrow 0$ and $x_0 \rightarrow q_0 z_0$, which is bounded; hence $\frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2}$ tends to 1 and $\partial \ln u_I / \partial \ln \tilde{q} \rightarrow -2 < 0$. Continuity and strict monotonicity give a unique interior root of (66). $\square$

Proposition 5. (Counterpart to Proposition 2) *There is positive assortative matching when ideas join incumbents: $\frac{\partial \tilde{q}^*}{\partial z_0} > 0$.*

Proof. Differentiate (77) with respect to $\ln z_0$. Using (74) for the first factor and, for the other term,

$$\frac{\partial}{\partial \ln z_0} \left[\frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \right] = - \frac{4A\tau\tilde{q}^2(\lambda_0^2 + 2A\lambda_0)x_0^2}{[(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2]^2} \left(\frac{q_0 z_0}{x_0} \right)^\eta < 0, \quad (80)$$

we obtain

$$\begin{aligned} \frac{\partial^2 \ln u_I}{\partial \ln \tilde{q} \partial \ln z_0} &= -2\eta b \left(\frac{q_0 \tilde{q}^b B}{x_0} \right)^\eta \left(\frac{q_0 z_0}{x_0} \right)^\eta \left[1 + \frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \right] \\ &\quad + \frac{8A\tau\tilde{q}^2(\lambda_0^2 + 2A\lambda_0)x_0^2}{[(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2]^2} \left(\frac{q_0 z_0}{x_0} \right)^\eta \left[1 - \frac{\partial \ln x_0}{\partial \ln \tilde{q}} \right]. \end{aligned} \quad (81)$$

Evaluate (81) at $\tilde{q} = \tilde{q}^*(z_0)$. The first line is strictly positive because $\eta < 0$, $b > 0$. The second line is strictly positive because, by Lemma 1, $\frac{\partial \ln x_0}{\partial \ln \tilde{q}} < \frac{1}{2} < 1$. Hence $\partial^2 \ln u_I / \partial \ln \tilde{q} \partial \ln z_0 > 0$: $\ln u_I$ is strictly supermodular in $(\ln \tilde{q}, \ln z_0)$.

Since $\partial^2 \ln u_I / \partial (\ln \tilde{q})^2 < 0$ at $\tilde{q}^*$ by (79), the implicit function theorem applied to (77) = 0 gives

$$\frac{d \ln \tilde{q}^*}{d \ln z_0} = - \frac{\partial^2 \ln u_I / \partial \ln \tilde{q} \partial \ln z_0}{\partial^2 \ln u_I / \partial (\ln \tilde{q})^2} \bigg|_{\tilde{q}=\tilde{q}^*} > 0, \quad \text{hence} \quad \frac{\partial \tilde{q}^*}{\partial z_0} > 0. \quad (82)$$

$\square$

Lemma 3. Write $x_0^* \equiv x_0(z_0, \tilde{q}^*(z_0))$. Then

$$u_I(z_0) = \frac{\lambda_0^3}{2(\lambda_0 + 2A)} \cdot \frac{1 - 2\frac{\partial \ln x_0}{\partial \ln \tilde{q}}}{1 - \frac{\partial \ln x_0}{\partial \ln \tilde{q}}} \cdot x_0^{*2} \nu Q, \quad u_N(z_0) = \frac{\lambda_0^3}{2(\lambda_0 + 2A)} z_0^2 \nu Q. \quad (83)$$

Proof. For a startup, $\tilde{q} = 0$ and $x_0 = z_0$, so (75) gives $a^* = \lambda_0 / (\lambda_0 + 2A)$ and (76) gives $u_N = \frac{1}{2} \frac{\lambda_0}{\lambda_0 + 2A} \lambda_0^2 z_0^2 \nu Q$, which is the expression in the baseline model.

For an incumbent, the utility function is:

$$u(z_0, \tilde{q}) = \frac{1}{2} a^* \lambda_0^2 x_0^2 \nu Q = \frac{\lambda_0^4}{2} \cdot \frac{x_0^4}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \nu Q. \quad (84)$$

The first-order condition can be rearranged as

$$\begin{aligned} \frac{2A\tau\tilde{q}^{*2}}{(\lambda_0^2 + 2A\lambda_0)x_0^{*2} + 2A\tau\tilde{q}^{*2}} &= \frac{\frac{\partial \ln x_0}{\partial \ln \tilde{q}}}{1 - \frac{\partial \ln x_0}{\partial \ln \tilde{q}}}. \\ \frac{(\lambda_0^2 + 2A\lambda_0)x_0^{*2}}{(\lambda_0^2 + 2A\lambda_0)x_0^{*2} + 2A\tau\tilde{q}^{*2}} &= \frac{1 - 2\frac{\partial \ln x_0}{\partial \ln \tilde{q}}}{1 - \frac{\partial \ln x_0}{\partial \ln \tilde{q}}}. \end{aligned}$$

Substituting into (76) and using $\lambda_0^2 + 2A\lambda_0 = \lambda_0(\lambda_0 + 2A)$ delivers (83). $\square$

Lemma 4. *If $b < 1$, then for every $z_0 > 0$*

$$\frac{d}{d \ln z_0} \left[\ln u_I(z_0) - \ln u_N(z_0) \right] = \frac{2(b-1) \left(\frac{q_0 \tilde{q}^{*b} B}{x_0^*} \right)^\eta}{1 - b \left(\frac{q_0 \tilde{q}^{*b} B}{x_0^*} \right)^\eta} < 0. \quad (85)$$

Proof. Because $\tilde{q}^*(z_0)$ maximizes $u_I(z_0, \cdot)$, the envelope theorem gives $d \ln u_I(z_0) / d \ln z_0 = \partial \ln u_I(z_0, \tilde{q}) / \partial \ln z_0$ evaluated at $\tilde{q}^*$. Taking logs in (76) and differentiating with respect to $\ln z_0$, using (71),

$$\frac{\partial \ln u_I}{\partial \ln z_0} = 2 \frac{\partial \ln x_0}{\partial \ln z_0} \left[1 + \frac{2A\tau\tilde{q}^2}{(\lambda_0^2 + 2A\lambda_0)x_0^2 + 2A\tau\tilde{q}^2} \right]. \quad (86)$$

At $\tilde{q} = \tilde{q}^*$ the FOC gives that:

$$\frac{d \ln u_I(z_0)}{d \ln z_0} = \frac{2 \frac{\partial \ln x_0}{\partial \ln z_0}}{1 - \frac{\partial \ln x_0}{\partial \ln \tilde{q}}}. \quad (87)$$

Given that

$$\frac{\partial \ln x_0}{\partial \ln z_0} = (q_0 z_0 / x_0^*)^\eta = 1 - (q_0 \tilde{q}^{*b} B / x_0^*)^\eta = 1 - \frac{1}{b} \frac{\partial \ln x_0}{\partial \ln \tilde{q}}, \quad (88)$$

the equation can be written as:

$$\frac{d [\ln u_I - \ln u_N]}{d \ln z_0} = \frac{2 \left[1 - \left(\frac{q_0 \tilde{q}^{*b} B}{x_0^*} \right)^\eta \right] - 2 \left[1 - b \left(\frac{q_0 \tilde{q}^{*b} B}{x_0^*} \right)^\eta \right]}{1 - b \left(\frac{q_0 \tilde{q}^{*b} B}{x_0^*} \right)^\eta} = \frac{2(b-1) \left(\frac{q_0 \tilde{q}^{*b} B}{x_0^*} \right)^\eta}{1 - b \left(\frac{q_0 \tilde{q}^{*b} B}{x_0^*} \right)^\eta}. \quad (89)$$

The numerator is negative for $b < 1$ and the denominator is positive by Lemma 1. $\square$

Proposition 6. (Counterpart to Proposition 3) When $b < 1, q_0 > 1$, there exists a cutoff $\bar{z}_0(B)$, such that all inventors with $z_0 < \bar{z}_0(B)$ join incumbent firms.

Proof. The inventor joins an incumbent iff $u_I(z_0) > u_N(z_0)$. By Lemma 4, $\ln u_I(z_0) - \ln u_N(z_0)$ is continuous and strictly decreasing on $(0, \infty)$. It remains to sign its limits.

From equation 77, it is straightforward that when $z_0 \rightarrow 0, \frac{z_0}{q^{*b}} \rightarrow 0$.

(i) $z_0 \rightarrow 0^+$. With $\eta < 0, z_0^\eta \rightarrow +\infty$, so

$$\frac{\partial \ln x_0}{\partial \ln \tilde{q}} = b(q_0 \tilde{q}^{*b} B / x_0)^\eta = b \frac{(q_0 \tilde{q}^{*b} B)^\eta}{z_0^\eta + (\tilde{q}^{*b} B)^\eta} \rightarrow 0 \quad (90)$$

$$x_0^* = q_0(z_0^\eta + (\tilde{q}^{*b} B)^\eta)^{1/\eta} \rightarrow q_0 z_0 \quad (91)$$

$$u_I(z_0) = \frac{\lambda_0^3}{2(\lambda_0 + 2A)} \cdot \frac{1 - 2 \frac{\partial \ln x_0}{\partial \ln \tilde{q}}}{1 - \frac{\partial \ln x_0}{\partial \ln \tilde{q}}} \cdot x_0^{*2} \nu Q \rightarrow \cdot q_0^2 u_N(z_0) \quad (92)$$

whenever $q_0 > 1$: inventors with ordinary ideas strictly prefer incumbents, whose commercialization scale is q_0 .

(ii) $z_0 \rightarrow \infty$. Suppose $\ln u_I - \ln u_N$ were bounded below. Being strictly decreasing it would converge to a finite limit, so along a subsequence its derivative (85) would tend to 0; since $b < 1$, this forces $(q_0 \tilde{q}^{*b} B / x_0^*)^\eta \rightarrow 0$, which by the argument in (i) implies $\ln u_I - \ln u_N \rightarrow 2 \ln q_0$. That contradicts strict monotonicity, since $\ln u_I - \ln u_N < 2 \ln q_0$ for every $z_0 > 0$. Hence $\ln u_I - \ln u_N \rightarrow -\infty$.

By continuity and strict monotonicity there is a unique $\bar{z}_0(B)$ with $u_I(\bar{z}_0) = u_N(\bar{z}_0)$, with $u_I > u_N$ below it and $u_I < u_N$ above it. $\square$

Proposition 7. (Counterpart to Proposition 4) When $b < 1, q_0 > 1$, if z_0 is drawn from a distribution $\Psi_{\text{new}}(z_0)$ that first-order stochastically dominates $\Psi(z_0)$, so ideas are better on average, a larger share of inventors launch startups.

Proof. By Proposition 6 the occupational choice is a threshold rule in own idea quality: inventor z_0 chooses a startup if and only if $z_0 > \bar{z}_0(B)$. The threshold does not depend on the idea distribution.

Let $\Psi_{\text{new}}(z_0)$ first-order stochastically dominate $\Psi(z_0)$. The share of inventors who launch startups is $h(1 - \Psi(\bar{z}_0(B)))$, and first-order stochastic dominance implies $\Psi_{\text{new}}(\bar{z}_0(B)) \leq \Psi(\bar{z}_0(B))$. Hence

$$1 - \Psi_{\text{new}}(\bar{z}_0(B)) \geq 1 - \Psi(\bar{z}_0(B)), \quad (93)$$

with strict inequality whenever the dominance is strict at $\bar{z}_0(B)$: when ideas are better on average, a larger share of inventors start new businesses. $\square$

C.12 Nash Bargaining

Suppose instead that each inventor–firm pair splits the match surplus through Nash bargaining with a common inventor weight $\theta \in (0, 1]$. Writing the inventor’s utility and the firm’s profit from a match as

$$\begin{aligned} U_I &= \left[a\lambda_0 e_I x_0(z_0, \tilde{q}) + T_I - Aa^2(\tau\tilde{q}^2 + \lambda_0 x_0(z_0, \tilde{q})^2) - \frac{1}{2}e_I^2 \right] \nu Q, \\ \Pi_F &= [(1-a)\lambda_0 e_I x_0(z_0, \tilde{q}) - T_I] \nu Q, \end{aligned}$$

the joint surplus $S \equiv U_I + \Pi_F = s(z_0, \tilde{q}; a) \nu Q$ is independent of the transfer T_I , since T_I cancels. The contract solves

$$\max_{a, T_I} (U_I - \bar{u})^\theta \Pi_F^{1-\theta} \quad \text{s.t.} \quad e_I = \lambda_0 a x_0(z_0, \tilde{q}).$$

Because T_I transfers utility one-for-one, the first-order conditions separate into an efficiency condition and a division condition:

$$a^* = \arg \max_a s(z_0, \tilde{q}; a), \quad U_I - \bar{u} = \theta (S - \bar{u}), \quad \Pi_F = (1 - \theta)(S - \bar{u}).$$

The equity share solves the same surplus-maximization problem as before, so

$$a^* = \frac{1}{1 + 2\frac{A}{\lambda_0} \left(\frac{\tau\tilde{q}^2}{\lambda_0 x_0(z_0, \tilde{q})^2} + 1 \right)}, \quad e_I = \lambda_0 a^* x_0(z_0, \tilde{q}),$$

are unchanged, and θ affects only the wage T_I . Reviewing all offers, an inventor chooses her firm to maximize $U_I = \bar{u} + \theta(s(z_0, \tilde{q}; a^*)\nu Q - \bar{u})$; since θ and νQ are common across firms, this is a positive affine transformation of the joint surplus, so the optimal firm size $\tilde{q}^*(z_0)$ and the startup-versus-incumbent cutoff coincide with those in the Bertrand case. The Bertrand economy is the limit $\theta = 1$, in which the inventor’s outside option is bid up to the full match surplus and firms retain no innovation rents. Hence effort, equity, and the allocation of ideas are invariant to a uniform division of the surplus; only the level of the transfer, and with it the share of rents accruing to incumbents, depends on θ .

D Innovation value in an incumbent firm

D.1 Model

When an inventor with idea quality z_0 works in an incumbent firm with quality $\tilde{q}$, the resulting innovation value $\tilde{x}(z_0, \tilde{q})$ is a random draw from a uniform distribution $U((1 - \phi) x_0(z_0, \tilde{q}) \nu, (1 + \phi) x_0(z_0, \tilde{q}) \nu)$. The mean value of the innovation depends on x_0 , which takes the following functional form:

$$x_0(z_0, \tilde{q}) = \left(\frac{\tilde{q}}{\tilde{q}_0} \right)^b (B^\eta + z^\eta)^{\frac{1}{\eta}}$$

where the parameter η governs the elasticity of substitution between technology stock B and idea quality z . From the model, the economic value of a patent $\tilde{x}$ satisfies:

$$\frac{\tilde{x}}{\tilde{q}^b} = \varepsilon (B^\eta + z^\eta)^{\frac{1}{\eta}}$$

where $\varepsilon = \frac{\nu}{\tilde{q}_0^b} \epsilon$. ν is an equilibrium outcome, $\tilde{q}_0$ and b are parameters. ϵ is stochastic.

D.2 Data

The economic value of a patent is estimated by [Kogan et al. \(2017\)](#). The firm size is mapped to the employment size.³⁶ B and z are measured using average backward citations in a year and the number of forward citations received by each patent.³⁷

D.3 Calibration

We estimate the parameter η using maximum likelihood estimation (MLE). To simplify calculation, assume that ε follows a log-normal distribution $\log(\varepsilon) \sim N(\mu_\varepsilon, \sigma_\varepsilon^2)$.³⁸ For each observation, the probability to happen is:

$$Pr(\tilde{x}, \tilde{q}, b, B, z; \eta, \mu_\varepsilon, \sigma_\varepsilon^2) = \frac{1}{\sigma_\varepsilon \sqrt{2\pi}} \exp \left\{ -\frac{\left(\log \left(\frac{\tilde{x}}{\tilde{q}^b} \right) - \frac{1}{\eta} (B^\eta + z^\eta) - \mu_\varepsilon \right)^2}{2\sigma_\varepsilon^2} \right\}$$

³⁶The firm size can also be measured using sales. The estimation is similar.

³⁷Both are truncated at 5-year window as before

³⁸If we adopt the uniform assumption, the results will be similar

Choose $\{\eta, \mu_\varepsilon, \sigma_\varepsilon^2\}$ to maximize the aggregate likelihood. The estimation result yields:

$$\eta = -0.31$$

Meaning that the elasticity of substitution is $\frac{1}{1+0.4} = 0.71$.

E Quantification Details

E.1 Distribution: Matching the Model with Data

The distributional data are from the BDS, which provides information on the number of firms and total employment in each size bin, capped at 10,000 employees per firm. We use a spliced log-normal distribution with a Pareto tail to parametrically represent the firm size distribution in the initial period. Specifically, the distribution is assumed to be log-normal when $q < \bar{q}$ and Pareto otherwise. The cutoff is set at 1,000 employees.³⁹ The parameters are estimated using GMM. Moments are divided into two sets:

1. cumulative density related, $g_{cdf,i}$, and
2. average employment related, $g_{emp,j}$.

The first type of moments measures the cumulative density in each bin, while the second type matches the average employment of firms within each bin. With 10 bins, this yields 19 moments to match 4 parameters.

Numerically, the estimated distribution is used as the initial distribution. The main challenge is converting from a per-employee measure to the model measure, which is expressed in terms of Q . We proceed with the following algorithm:

1. Calculate the measure of firms, N_F , in the initial period.
2. Construct a grid of employment levels covering both the log-normal and Pareto parts of the distribution.
3. Calculate the CDF from the parametric distribution.
4. Using the CDF, compute the mass of firms in each bin, ensuring that the masses sum to the total measure of firms.
5. Calculate Q based on the grids and the mass allocation.
6. Normalize the grid using Q .

The normalized grid, together with the mass from step 4, is then used as the initial distribution. When simulating, the normalization is re-set every instant. As the simulation runs, it moves forward in very small increments of time, dt . The simulation advances in small increments dt , and the reference grid is renormalized at each step, so the distribution evolves smoothly rather than jumping from year to year.

³⁹We test alternative cutoffs, which do not materially affect the results.

E.2 Compute HHI

We compute the HHI from the simulated firm distribution defined on the discrete state grid. Our baseline measure is obtained directly from the model-implied cross-sectional distribution. Let q_{it} denote total output associated with grid point i at time t . We define $HHI_t = \sum_i (q_{it} / \sum_i q_{it})^2$. This measure corresponds to the model-implied population concentration.

As a numerical validation, we alternatively compute the HHI using a large Monte Carlo sample. In each period t , we draw firms in proportion to the total firm measure N_{Ft} and calculate the implied concentration. By the law of large numbers, the sample-based HHI converges to the population counterpart. Quantitatively, the two procedures yield indistinguishable results.

Because firms are modeled as a continuum, the level of the HHI depends mechanically on how we discretize the firm-size state space and therefore has no standalone economic interpretation. Accordingly, we focus on trends and fluctuations, reporting the HHI normalized by its mean and emphasizing its time-series dynamics.

E.3 Robustness Checks—Different Initial Year

The following figures plot the simulated growth rates in each 5-year window, annual HHI and its extensive-margin component, and the new-to-incumbent ratio when the simulation starts in 1986. The results closely match the simulation starting from 1984.

E.4 Growth Decomposition—Simulation with Variation in z_m Only

Figure 19 decomposes simulated growth into growth in average firm quality and changes in firm mass when only z_m varies over time.

E.5 Details of the HHI Counterfactual

This appendix describes the counterfactual used to isolate the intensive-margin contribution to market concentration.

1. *Optimal incumbent size.* For every inventor with idea quality z_0 , we solve for the optimal incumbent size $\tilde{q}^*(z_0, t)$ conditional on developing the idea inside an incumbent, exactly as in the baseline.
2. *Occupational choice.* Each inventor then compares the utility of joining that incumbent with the utility of founding a startup, using the same contracts and

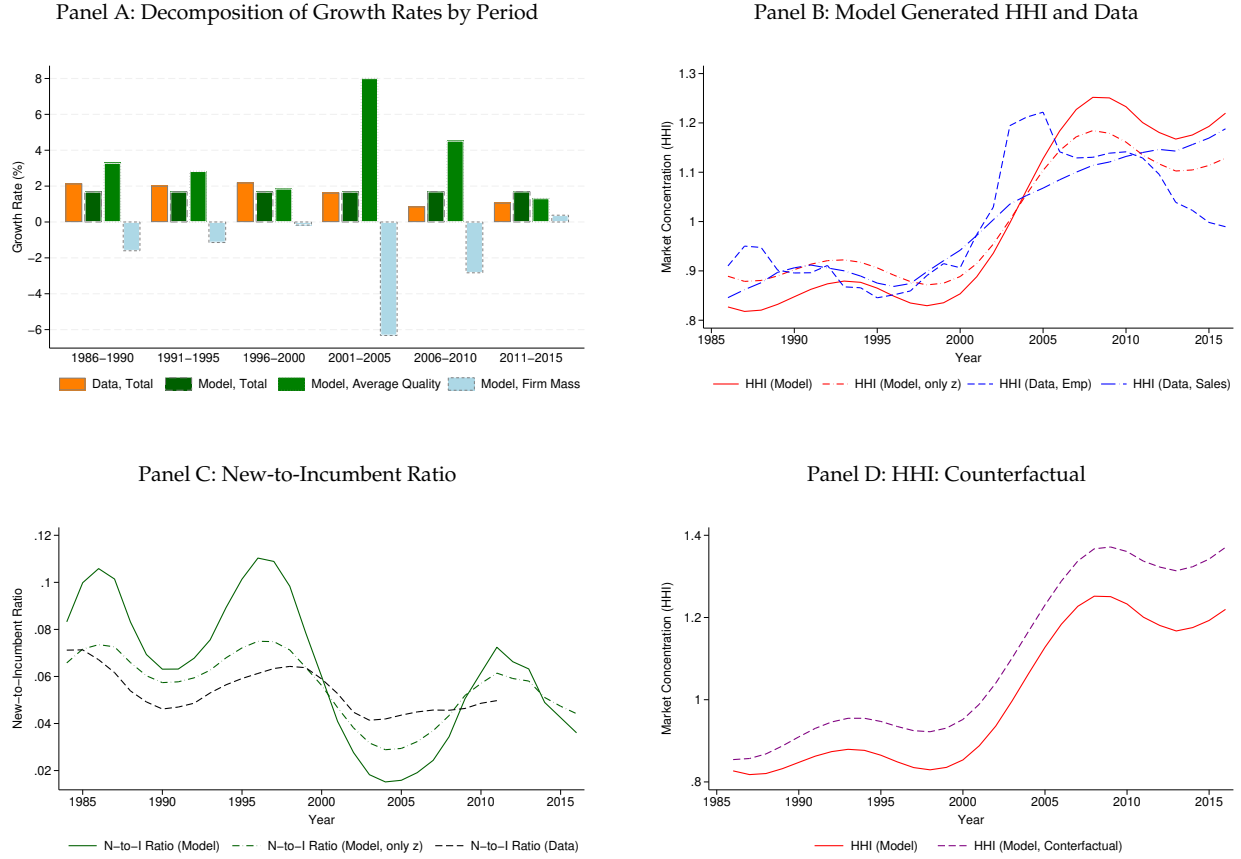


Figure 18: Simulation Results Starting from 1986

Notes: Panel A reports five-year average aggregate growth and decomposes it into improvements in firm quality and changes in firm mass. Panel B compares normalized HHI over time in the model (simulated annual HHI) and in the data (employment-based HHI from the LBD and sales-based HHI from Compustat). Panel C plots the new-to-incumbent ratio in both the model and the data. Panel D isolates the contribution of the intensive margin by comparing the full-model HHI with the counterfactual HHI. All simulations begin in 1986.

the same comparison as in equation (24). The cutoff $\bar{z}_0$ is therefore identical to the baseline, and so is the set of inventors who found startups. Therefore, the extensive margin is preserved.

3. *Inventors who join incumbents.* For these inventors nothing changes. The equity share, effort, and the realized quality improvement $x_0(z_0, \tilde{q}^*)$ are those of the baseline, and the idea is developed in the incumbent of size $\tilde{q}^*(z_0, t)$ that the inventor selects.
4. *Inventors who join startups.* These inventors innovate inside the startup and solve the same contracting problem as in the baseline. A new firm enters upon a successful innovation, and the entrant draws its relative size $\tilde{q}_I$ from the entry distribution

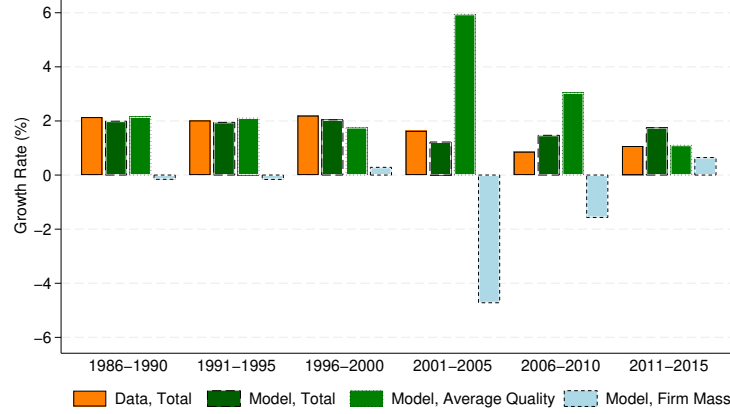


Figure 19: Growth Decomposition with Variation in z_m Only

Notes: The figure reports five-year average aggregate growth and decomposes it into growth in average firm quality and changes in firm mass. In this simulation, only the scale parameter of the idea-quality distribution, z_m , varies over time; the technology stock, B , and all other parameters are held fixed.

$f_{I,t}(\tilde{q}_I)$, exactly as in the baseline. Hence, the entry rate $\lambda_I(t)$ is unchanged, and so is the entire path of firm mass $N_F(t)$.

5. *Startup innovation reassignment.* The entrant enters at its drawn size $\tilde{q}_I$ only. The innovation created by the startup inventor is then transferred to an incumbent of size $\tilde{q}^*(z_0, t)$, the optimal size obtained in Step 1. Startup ideas are thus matched to firm size by the same assortative rule that governs incumbent ideas. The transferred step size is the raw idea quality z , with mean z_0 , and not the commercialization value $x_0(z_0, \tilde{q}^*) = (\tilde{q}^*/q_0)^b \gamma(z_0)$ that the idea would generate if it were genuinely developed inside that incumbent. The receiving incumbent is credited with neither the synergy factor $(\tilde{q}^*/q_0)^b$ nor the adoption friction embedded in $\gamma(\cdot)$. The exercise therefore excludes the synergy between inventors and firms.

E.6 IV Construction and First-Stage Results

Because the Tech Wave Index may be correlated with unobserved shocks to concentration and reverse causality may exist, we construct a shift-share instrument that combines the U.S. innovation network with lagged technology waves. Using U.S. patent data, we construct a time-varying diffusion matrix $\Omega_{c' \rightarrow c, t, \tau}$ at the IPC-class level that maps citation flows from an upstream (cited) class c' in year $t - \tau$ to a downstream (citing) class c in year t , for lags $\tau \in \{1, \dots, 5\}$. For each (c, t, τ) we expand patent citations to IPC $\times$ IPC pairs with fractional weights, drop within-class links, aggregate to (c', c, t, τ) , and normalize so that $\sum_{c' \neq c} \Omega_{c' \rightarrow c, t, \tau} = 1$. We then combine this diffusion structure with the Tech Wave Index at the IPC level, $TWI_{c', t - \tau}$, to form a shift-share instrument in the

spirit of [Acemoglu, Akcigit and Kerr \(2016\)](#) and [Liu et al. \(2025\)](#):

$$\lambda_{c,t} = \frac{1}{5} \sum_{\tau=1}^5 \sum_{c' \neq c} \Omega_{c' \rightarrow c, t, \tau} TWI_{c', t-\tau}.$$

Intuitively, $\lambda_{c,t}$ loads each downstream class on lagged breakthrough intensity in technologically connected upstream classes. The instrument is mapped from IPC to NAICS using the same concordance. Following the exogenous shocks framework of [Borusyak, Hull and Jaravel \(2022\)](#), our key identifying assumption is that the exact timing and magnitude of aggregate technological breakthroughs in the *upstream* classes (c') are quasi-random with respect to downstream (c), sector-specific unobservables. Because downstream market participants cannot dictate the fundamental scientific progress of their upstream suppliers, this network-based instrument isolates the effect of technological shocks and effectively mitigates the problem of reverse causality.

Table 12: First-stage Results of 2SLS Regressions

	HHI						New Firm Indicator	
	TWI _{s,t}		TWI _{s,t-1}		TWI _{s,t-2}		TWI _{c,t}	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\lambda_{s,t}$	0.571*** (0.157)	0.736*** (0.0874)						
$\lambda_{s,t-1}$			0.345* (0.183)	0.719*** (0.0849)				
$\lambda_{s,t-2}$					0.347* (0.184)	0.722*** (0.0864)		
$\lambda_{c,t}$							0.600*** (0.121)	0.707*** (0.0999)
Observations	37500	37500	37500	37500	36500	36500	4032000	4032000
Industry/IPC FE	No	Yes	No	Yes	No	Yes	No	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes
F excl. instruments	13.21	70.85	3.54	71.86	3.56	69.92	24.53	49.99

Notes: Standard errors are clustered at the 6-digit NAICS-by-year level for columns (1)–(6), and at the 4-digit IPC-by-year level for columns (7)–(8). The inclusion of year and industry fixed effects, and the F-statistics assessing instrument relevance are shown in the last three rows. To comply with Census Bureau disclosure requirements, the number of observations is rounded to the nearest hundred. *** Significant at the 1 percent level; ** significant at the 5 percent level; * significant at the 10 percent level.

Table 12 reports the first-stage results for the 2SLS regressions. Columns (1)–(2) relate the contemporaneous Tech Wave Index to its shift-share instrument in the HHI regression at the 6-digit NAICS-by-year level. Columns (3)–(4) and (5)–(6) report the corresponding first-stage relationships for the one-year and two-year lagged Tech Wave Index, respectively. Columns (7)–(8) show the first-stage relationship between the

contemporaneous Tech Wave Index and its instrument in the patent-level regression with the new-firm indicator as the outcome. The last row reports the first-stage F-statistics assessing instrument relevance. Across all specifications, these statistics indicate a strong and statistically significant relationship between the instrument and the endogenous regressor.